Christian-Albrechts-Universität zu Kiel
Intelligent Systems Lab

Data Drift Detection in Traversability Estimation for Safe Mobile Robot Navigation

MASTER'S THESIS

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Kiel, September 2026

Motivation

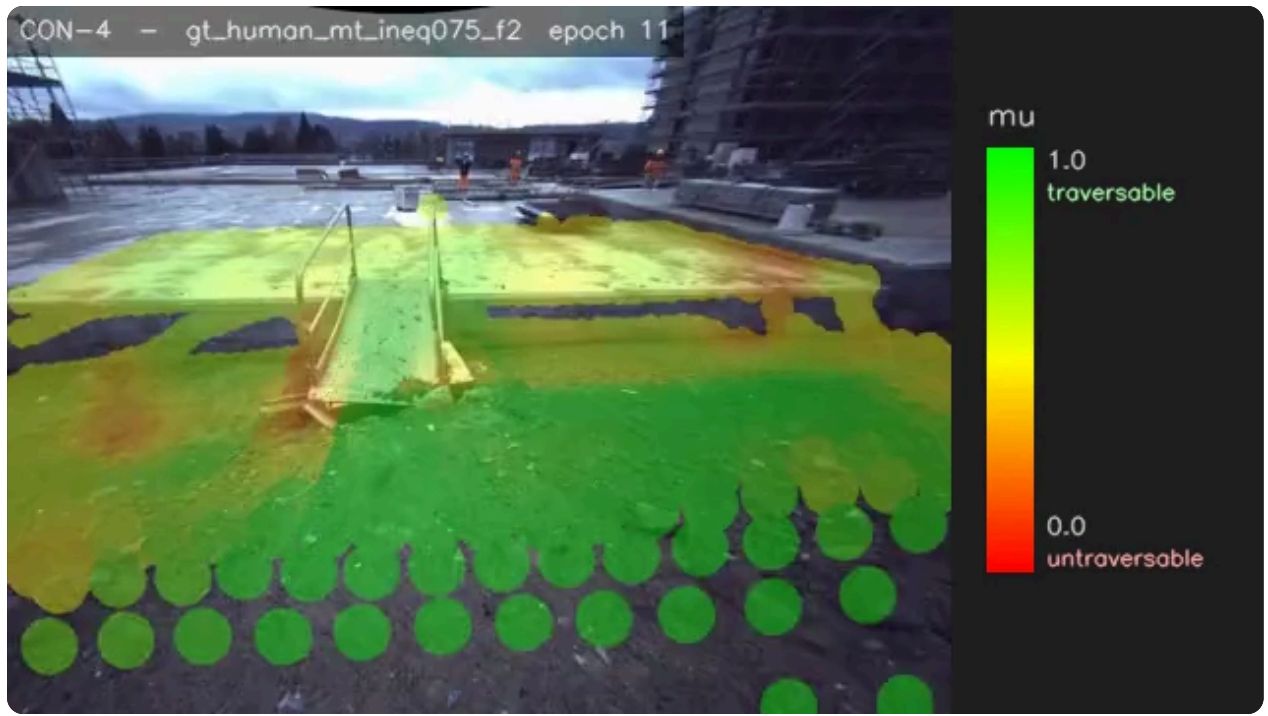
Traversability Estimation

Limitations of Geometric-only Approaches

- Over-conservative behavior in vegetation (e.g., tall grass)
- Deformable obstacles (e.g. snow, mud) and slippery surfaces may be falsely detected as traversable

Out-of-Distribution Detection

Goal: Detect data drift in the input distribution of a trained traversability estimator



CON-4, a construction site the estimator never trained on

THESIS CONCEPT

How can a self-supervised traversability estimation network be augmented to detect novel, out-of-distribution terrains?

Traversability Estimation + OOD Detection

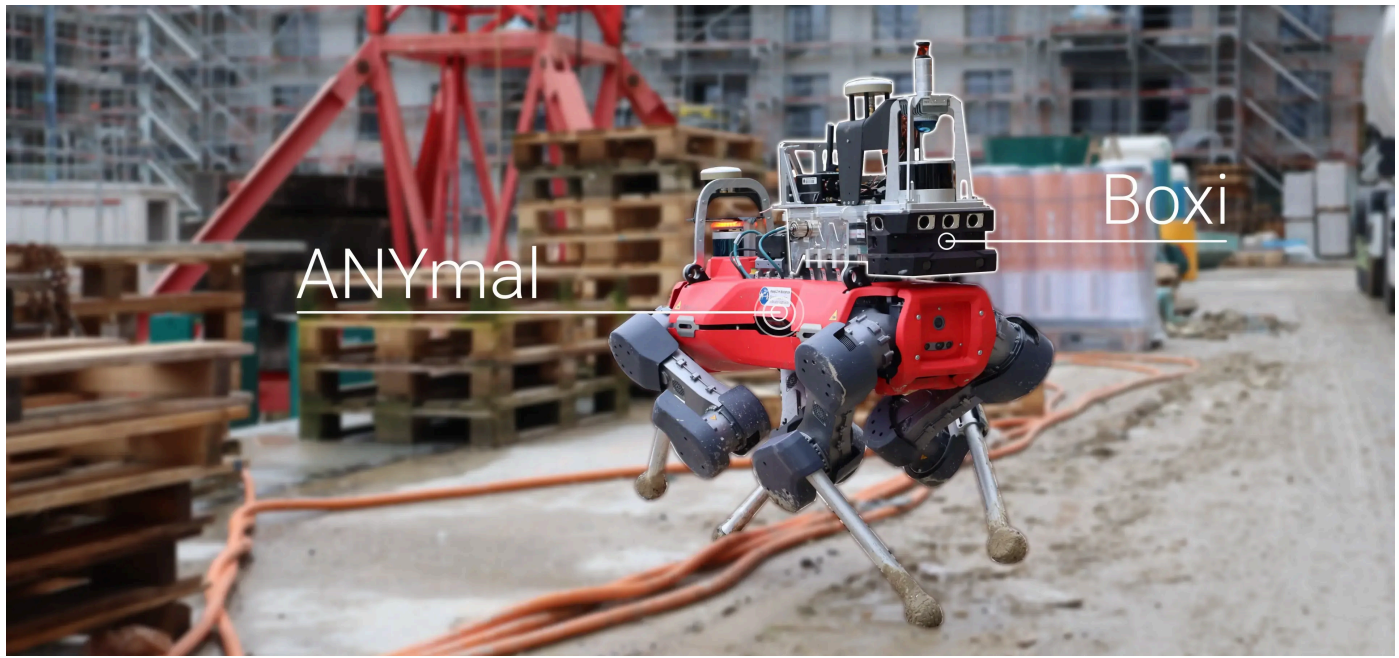
RQ1: WayFASTER on GrandTour

ARXIV:2602.18164 · 2026

GrandTour: A Legged Robotics Dataset in the Wild for Multi-Modal Perception and State Estimation

Jonas Frey, Turcan Tuna, Frank Fu, Katharine Patterson, Tianao Xu,
Maurice Fallon, Cesar Cadena, Marco Hutter

The GrandTour Dataset



ANYmal-D with the *Boxi* sensor payload

GrandTour: Frey et al., 2026.

The GrandTour Dataset

- Diverse environments, almost all at day-time
- Relatively large forest slice: 1h 40min

RQ1 · GRANDTOUR

Adapting the Dataset

Problem 1: Red Color Tint on Dark Parts of the Dataset

Problem 2: No Precise Odometry on Nine Missions

- Most missions have precise odometry information, some randomly do not
- DLIO LIDAR odometry
- Median deviation of 14 cm, compared to the toal station ground truth (where it exists)

Problem 3: No Usable Stereo Depth

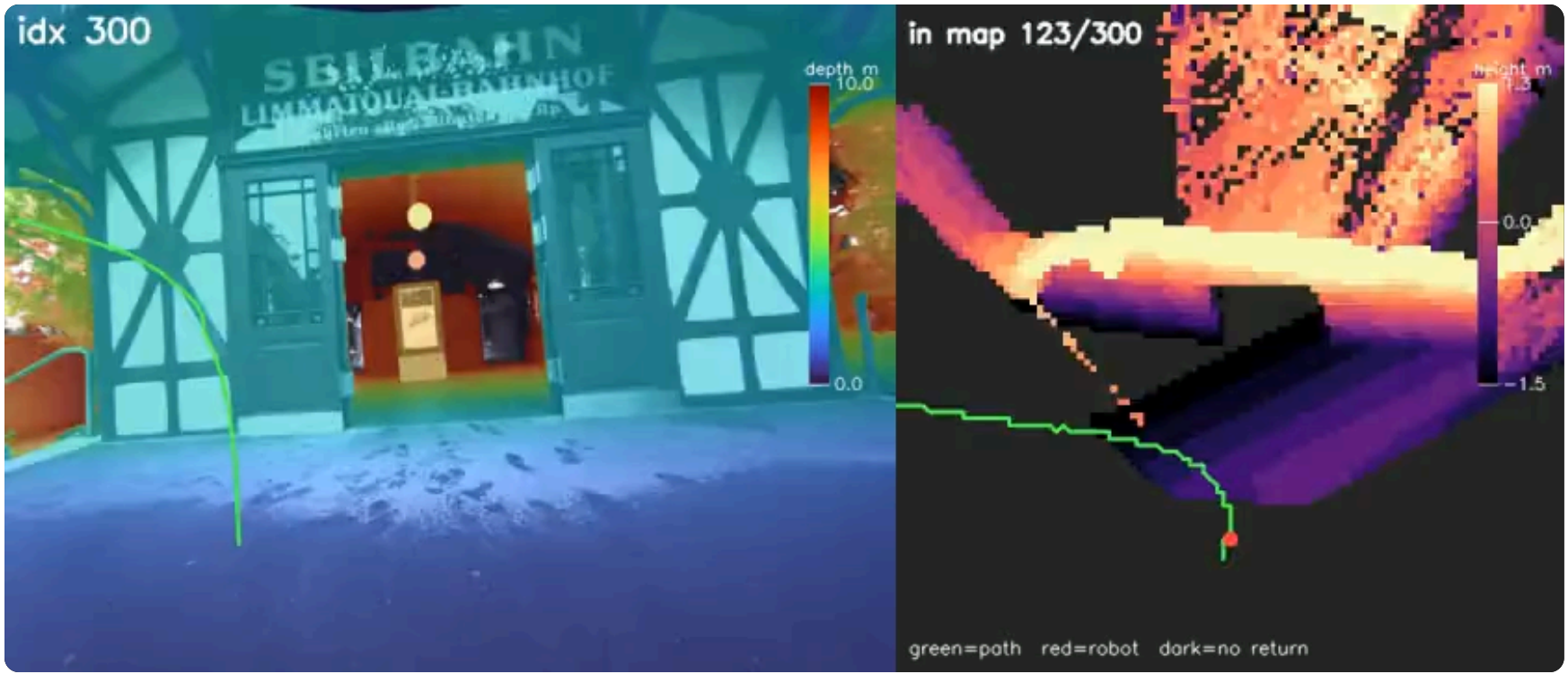
CVPR (ORAL), NVIDIA · 2025

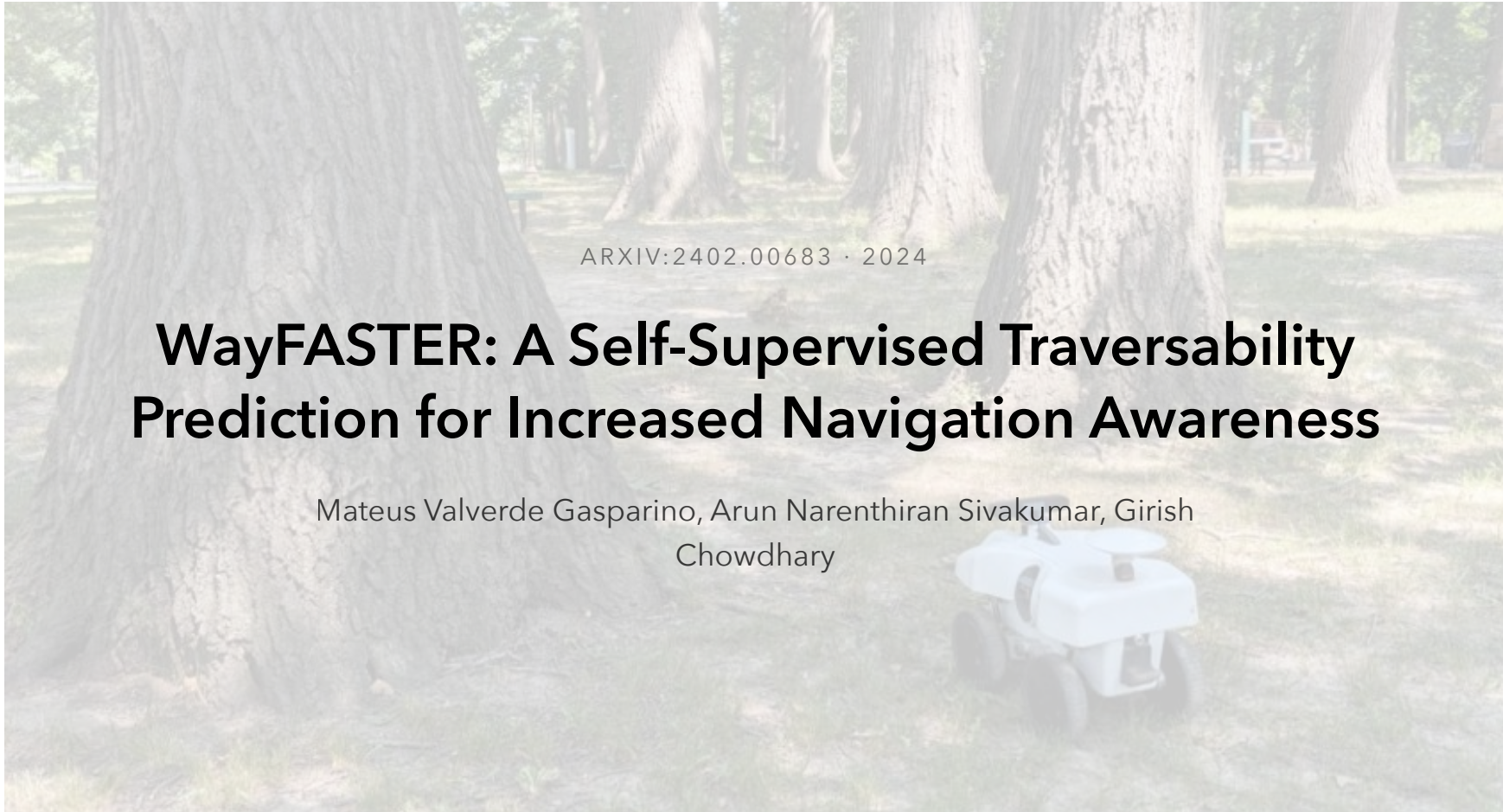
FoundationStereo: Zero-Shot Stereo Matching

Bowen Wen, Matthew Trepte, Joseph Aribido, Jan Kautz, Orazio Gallo, Stan Birchfield

Generate depth from stereo images.

Foundation Stereo: Visualization





ARXIV:2402.00683 · 2024

WayFASTER: A Self-Supervised Traversability Prediction for Increased Navigation Awareness

Mateus Valverde Gasparino, Arun Narenthiran Sivakumar, Girish Chowdhary

The WayFASTER Architecture

Rationale for the Architecture

- Accepted to ICRA'24 (largest robotics conference)
- Self-supervised labeling scheme
- Open source training code (but no labeling code)
- Consists of classic ML components (favourable for OOD detection)
- Real time inference on on-board hardware (NVIDIA Jetson Nano, etc.)

RQ1 · GRANDTOUR

Unsupervised Label Method: Receding Horizon Estimator (RHE)

The Kinematic Model

Position p_x, p_y

$$p_{x,k+1} = p_{x,k} + \mu\, v_k \cos\theta_k$$
$$p_{y,k+1} = p_{y,k} + \mu\, v_k \sin\theta_k$$

where:

- v_k commanded linear velocity
- μ scale factor on commanded linear velocity (**optimized**)

Heading Angle θ

$$\theta_{k+1} = \theta_k + \nu\, \omega_k$$

where:

- ω_k commanded angular velocity.
- ν scale factor on commanded angular velocity (**optimized**)

Receding Horizon Optimization (Simplified Version)

$$x_{k+1} = f(x_k, u_k;\ \mu, \nu)$$

where:

- f is the kinematic model from the previous slide
- $x_k = (p_{x,k}, p_{y,k}, \theta_k)$ the state
- $u_k = (v_k, \omega_k)$ the commanded velocities

$$(\mu_k, \nu_k) = \underset{\mu,\,\nu}{\arg\min} \sum_{i=k-T}^{k} \left\| x_i^{\text{real}} - \hat{x}_i \right\|^2$$

where:

- x_i^{real} the pose measured by odometry at step i
- $\hat{x}_i$ the pose recursively predicted by the kinematic model, from the start of the window

Ensuring Label Quality

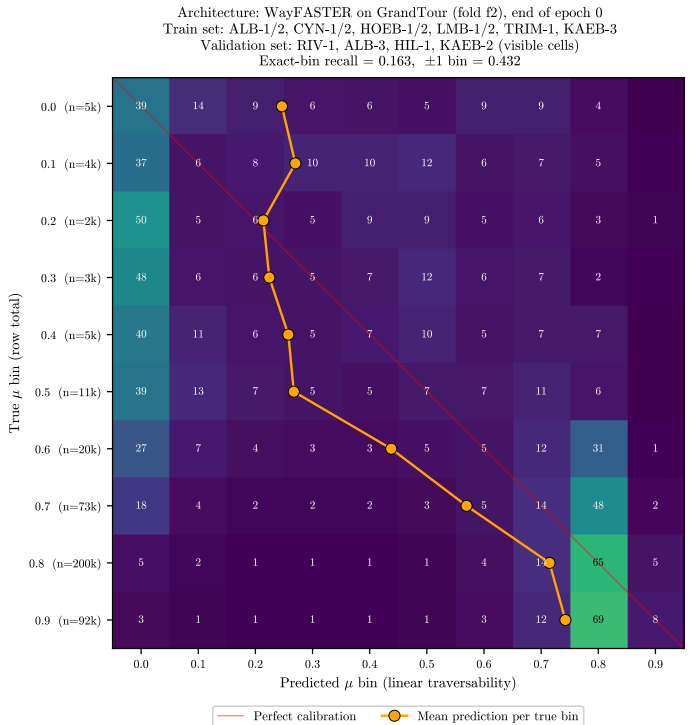
Why: If labels are bad, everything else is meaningless

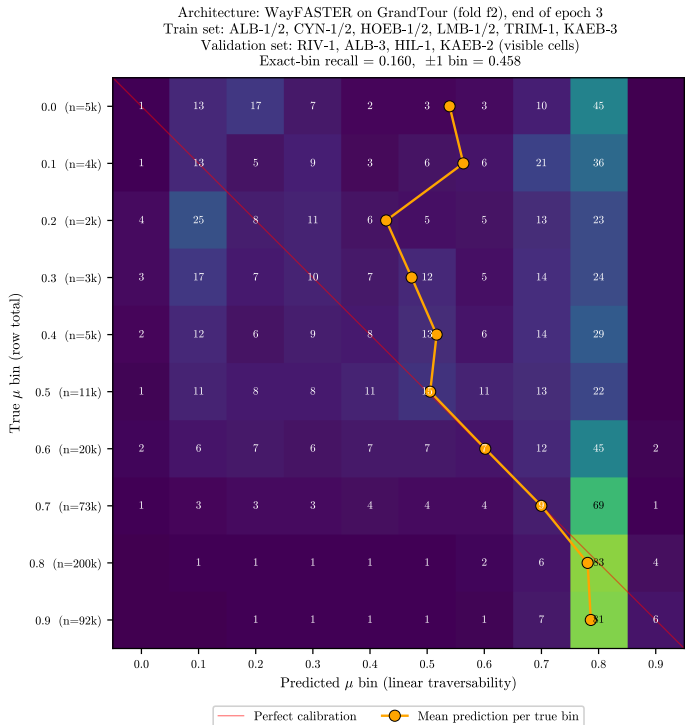
- Test on synthetic scenarios with known (μ, ν), recovered by the estimator
- RHE Hyperparameters tuned with an evolutionary algorithm
- Qualitatively check labels next to the traversed path
  - Labels look plausible, but noisy

RQ1 · GRANDTOUR

Training Results

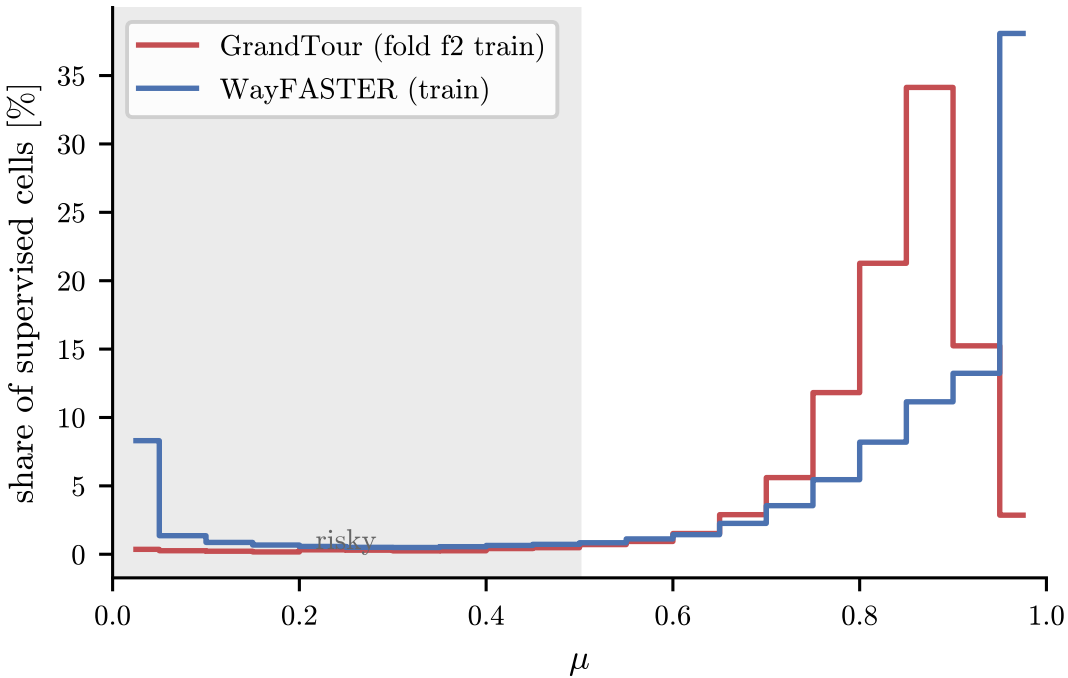
Training Results





Model collapses to the dataset mean after first or second epoch.

GrandTour Against WayFASTER



	GrandTour	WayFASTER
Recordings	10	47
Duration	1.1 h	7.1 h
Windows	7 759	50 721
Supervised cells	1.21 M	10.09 M
Cells with $\mu < 0.5$	3.0 %	14.7 %

Hypothesis

The combination of small dataset size and label sparsity makes it nearly impossible to learn patterns that generalize across recordings.

The Regularization Grid

Axis	Levels
Backbone	ResNet34 (ImageNet); DINOv3
Backbone plasticity	full LR; 0.1x LR; frozen
Decoder capacity	full; 1x1 `bev_compressor`, no `layer3`; minimal on frozen backbone; no μ decoder at all
Loss function	inverse-frequency weighted; BMC; Gaussian-binned CE; focal; WVN
Weight decay	10^{-4} to 10^{-1}
Depth-head weight	0.05 (baseline) to 0.3-0.8
Training data	ID only; ID + dead weight

Grid Results

	Max calib. slope ↑	Max AUROC ↑	Final slope ↑
Baseline (f2), four-seed mean	0.509	0.769	0.213
Noise floor (seed spread)	± 0.050	± 0.012	± 0.031
Backbone plasticity			
0.1x learning rate			
Frozen			
Decoder capacity			
Slim decoder			
Minimal, frozen backbone			
No μ decoder (linear on DINOv3)			
Loss function			
Balanced MSE			
Binned CE, two-hot			
Binned CE, Gaussian			
Focal-R			
W-RIZZ ordinal pairs			

	Max calib. slope ↑	Max AUROC ↑	Final slope ↑
Baseline (f2), four-seed mean	0.509	0.769	0.213
Noise floor (seed spread)	± 0.050	± 0.012	± 0.031
Weight decay			
0.01			
0.1			
Depth weight			
0.1			
0.5			
Training data			
All missions (4x data)			
WayFASTER pretraining			
Warm start			
Warm start, frozen encoder			

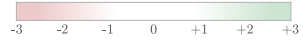
Distance from the baseline, in noise-floor standard deviations.
Boxed: best per column. ↑ higher is better.

Grid Results

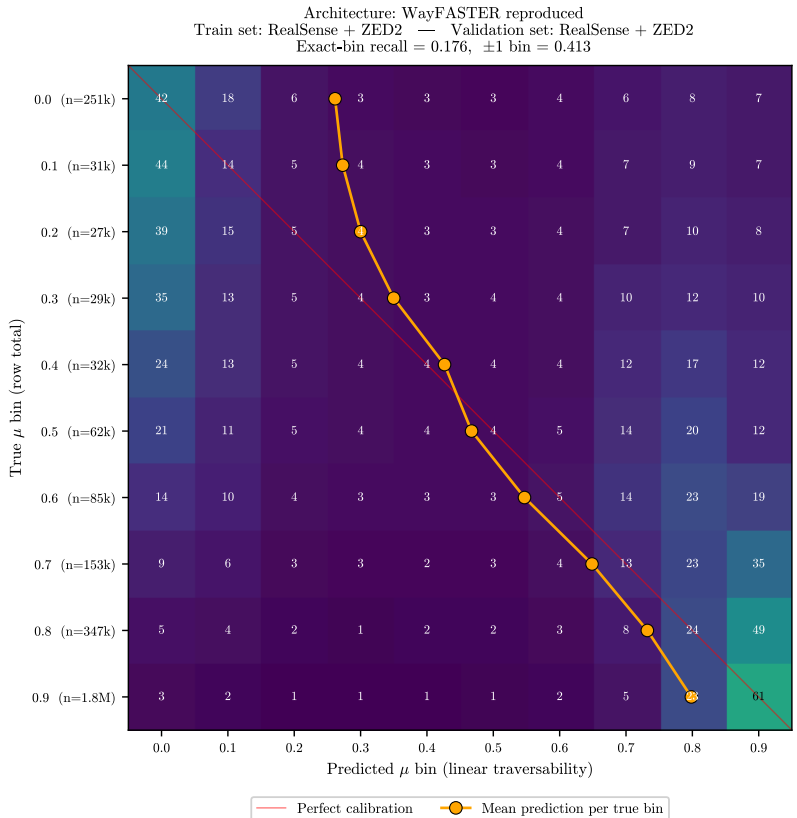
	Max calib. slope ↑	Max AUROC ↑	Final slope ↑
Baseline (f2), four-seed mean	0.509	0.769	0.213
Noise floor (seed spread)	± 0.050	± 0.012	± 0.031
Backbone plasticity			
0.1x learning rate	0.341	0.699	0.184
Frozen	0.445	0.735	0.223
Decoder capacity			
Slim decoder	0.397	0.715	0.218
Minimal, frozen backbone	0.298	0.679	0.225
No μ decoder (linear on DINOv3)	0.065	0.639	0.065
Loss function			
Balanced MSE	0.433	0.783	0.308
Binned CE, two-hot	0.368	0.769	0.185
Binned CE, Gaussian	0.300	0.748	0.123
Focal-R	0.470	0.779	0.176
W-RIZZ ordinal pairs	0.169	0.664	0.108

	Max calib. slope ↑	Max AUROC ↑	Final slope ↑
Baseline (f2), four-seed mean	0.509	0.769	0.213
Noise floor (seed spread)	± 0.050	± 0.012	± 0.031
Weight decay			
0.01	0.461	0.762	0.213
0.1	0.443	0.751	0.157
Depth weight			
0.1	0.460	0.764	0.244
0.5	0.521	0.783	0.311
Training data			
All missions (4x data)	0.562	0.777	0.321
WayFASTER pretraining			
Warm start	0.409	0.775	0.260
Warm start, frozen encoder	0.347	0.745	0.198

Distance from the baseline, in noise-floor standard deviations.
Boxed: best per column. ↑ higher is better.



Reproducing WayFASTER on the WayFASTER Dataset



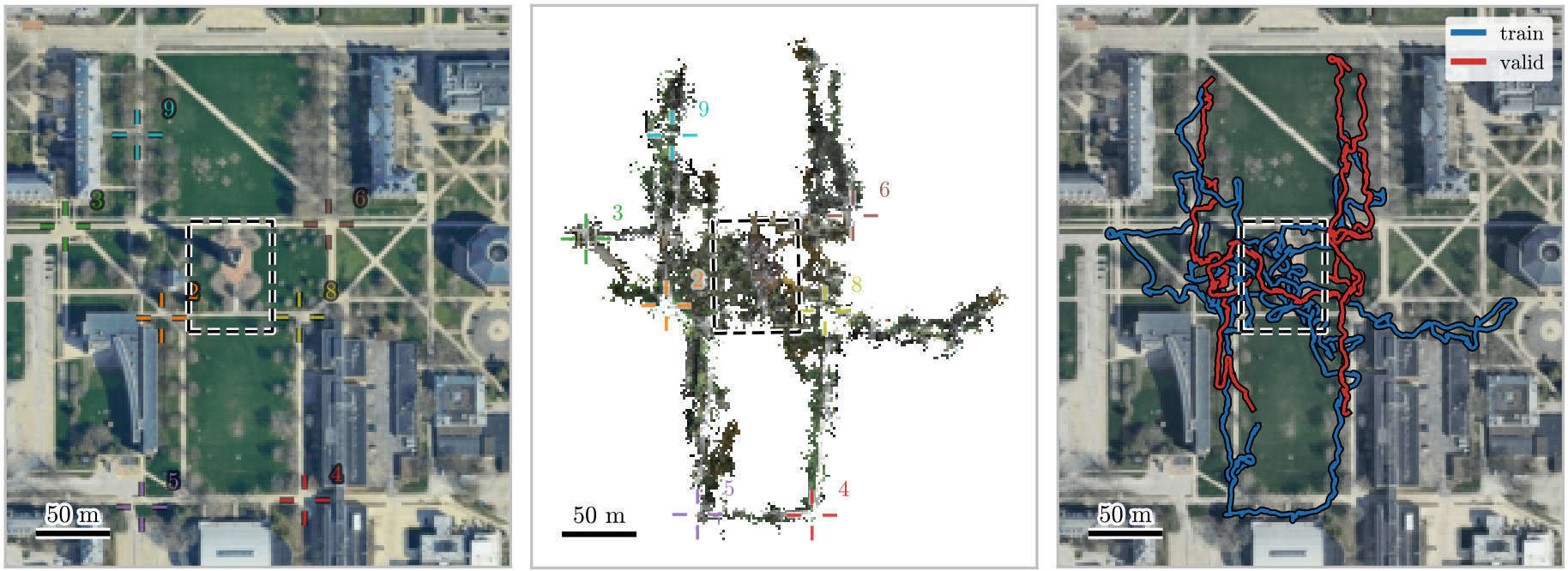
The WayFASTER Data Collection Strategy

Overlap Between Training and Validation

(a) Satellite map

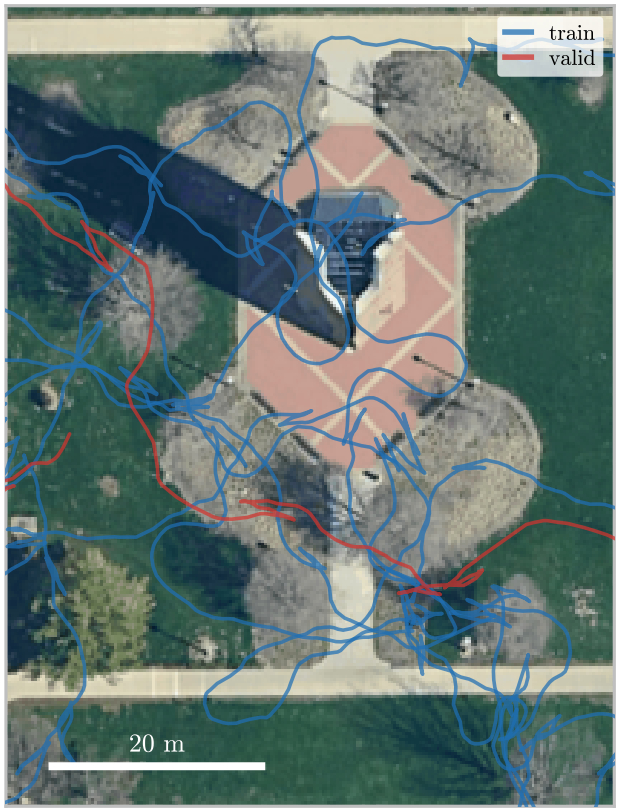
(b) Birds Eye View from the train and validation runs

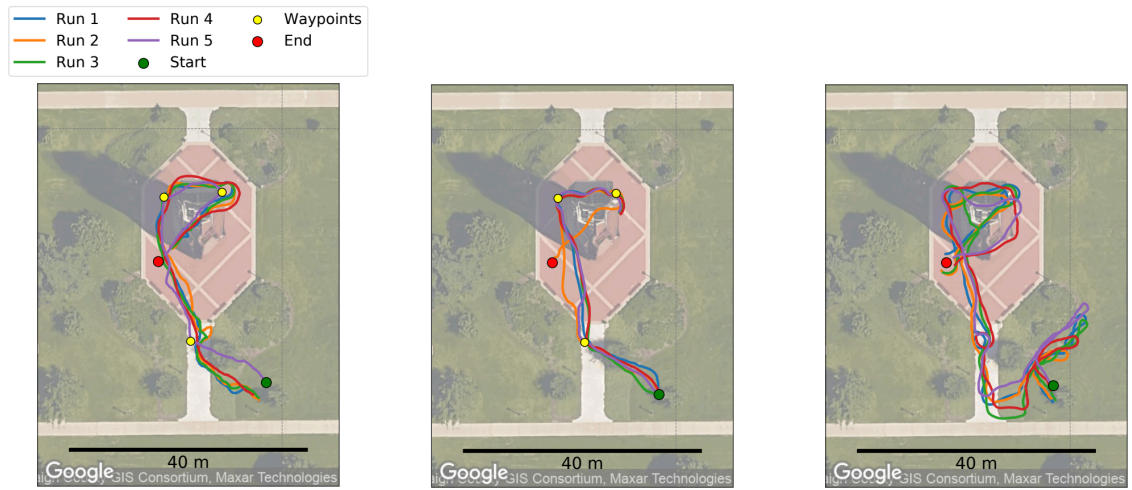
(c) Routes placed on the map



The `2023-06-28` slice, matched against satellite imagery.

Overlap with WayFASTER's Field Trials





Right: Fig. 4 of Gasparino et al., 2024.

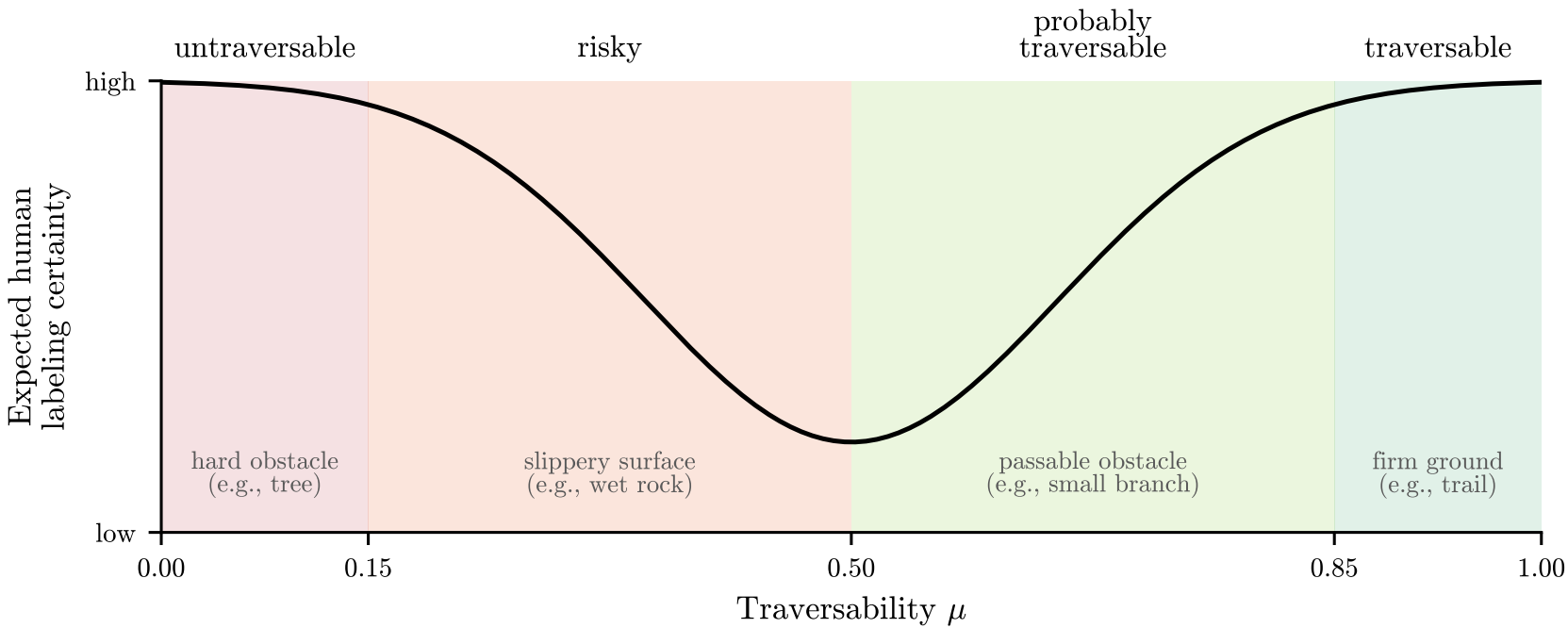
Relative Labeling

ARXIV:2406.02822 · 2024

W-RIZZ: A Weakly-Supervised Framework for Relative Traversability Estimation in Mobile Robotics

Andre Schreiber, Arun N. Sivakumar, Peter Du, Mateus V. Gasparino, Girish Chowdhary, Katherine Driggs-Campbell

Limits of Human and Proprioceptive Labels



W-RIZZ: Relative Labeling

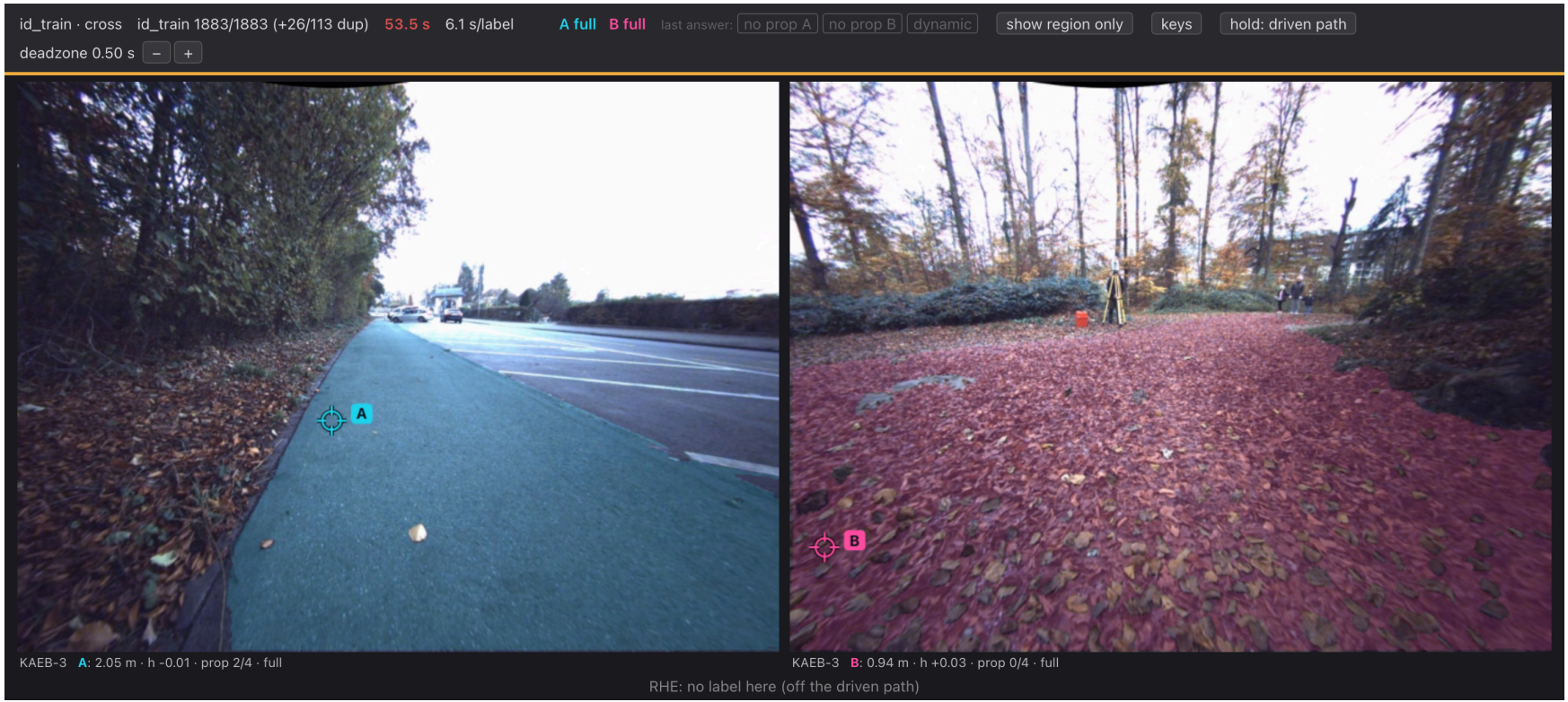
The W-RIZZ Loss

$$\mathcal{L}_{\text{RIZZ}}(p_a, p_b) = \begin{cases} \max\{0,\ L - (p_b - p_a)\}^2 & l_a < l_b \\ \max\{0,\ L - (p_a - p_b)\}^2 & l_a > l_b \\ (p_b - p_a)^2 & l_a = l_b \end{cases}$$

with:

- p_a, p_b the network's prediction at the two points
- l_a, l_b the annotator's answer
- $L = 0.5$ the margin.

Annotation Procedure

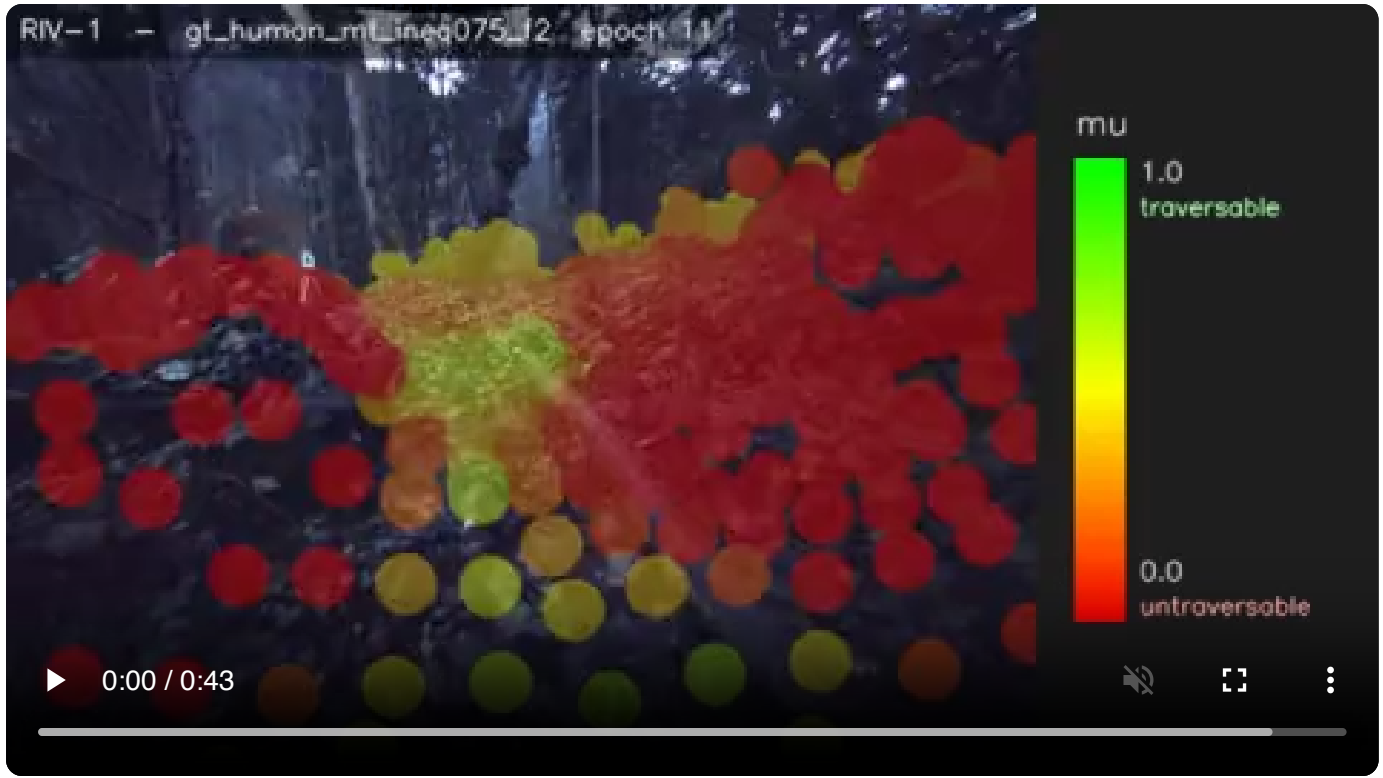


Human Disagreement Rate

$$\mathrm{HDR}_\tau = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\left(L_\tau(p_{i,a}, p_{i,b}) \neq t_i\right) \qquad L_\tau(p_a, p_b) = \begin{cases} +1 & p_b - p_a > \tau \\ 0 & |p_b - p_a| \leq \tau \\ -1 & p_b - p_a < -\tau \end{cases}$$

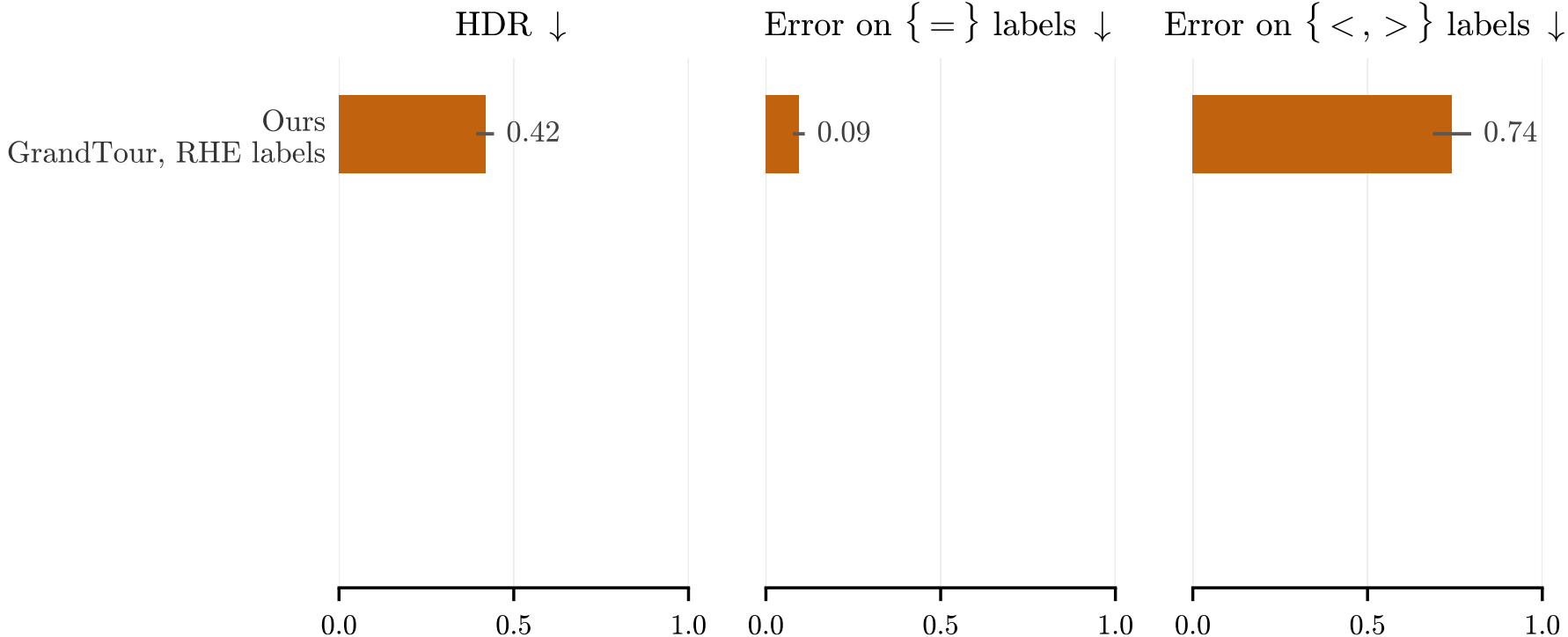
- $t_i \in \{-1, 0, +1\}$ is the annotator's answer
- $\tau = 0.25$ the deadband the paper reports at.

Results: Qualitative Check



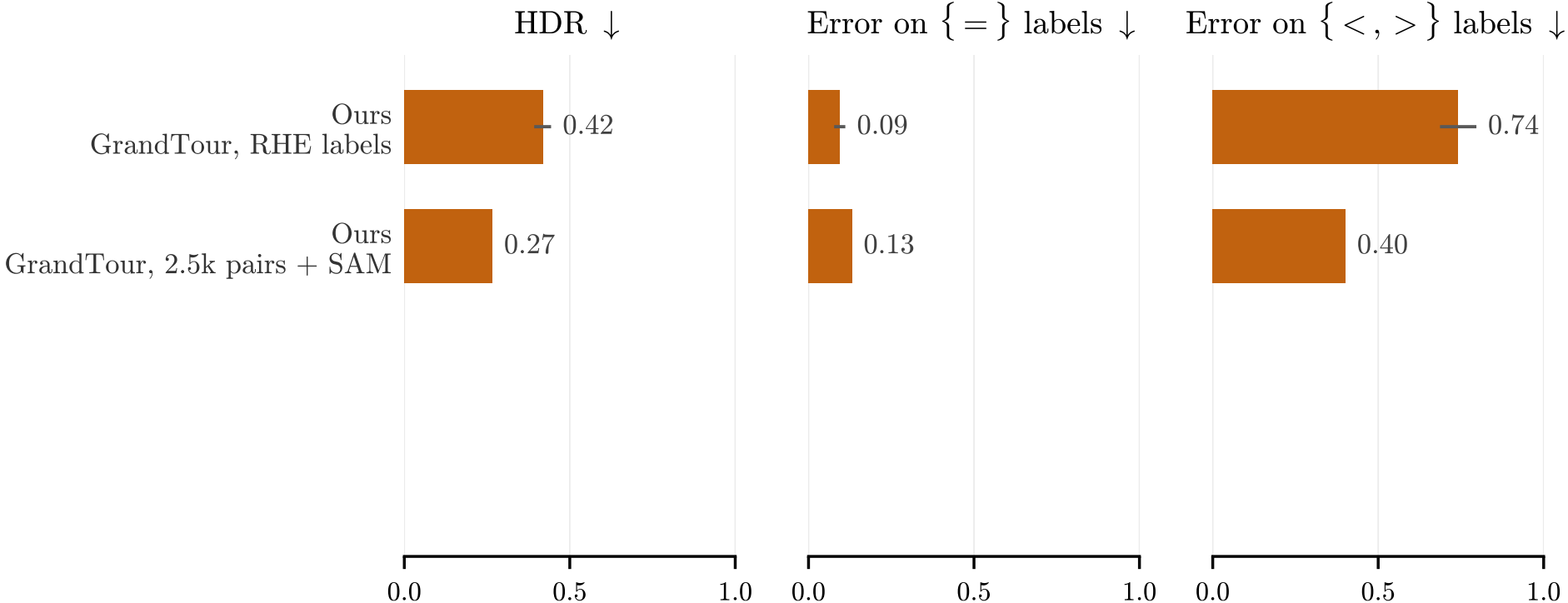
Trained on human labels, shown on the validation set: RIV-1, ALB-3, HIL-1, KAEB-2.

Results: HDR



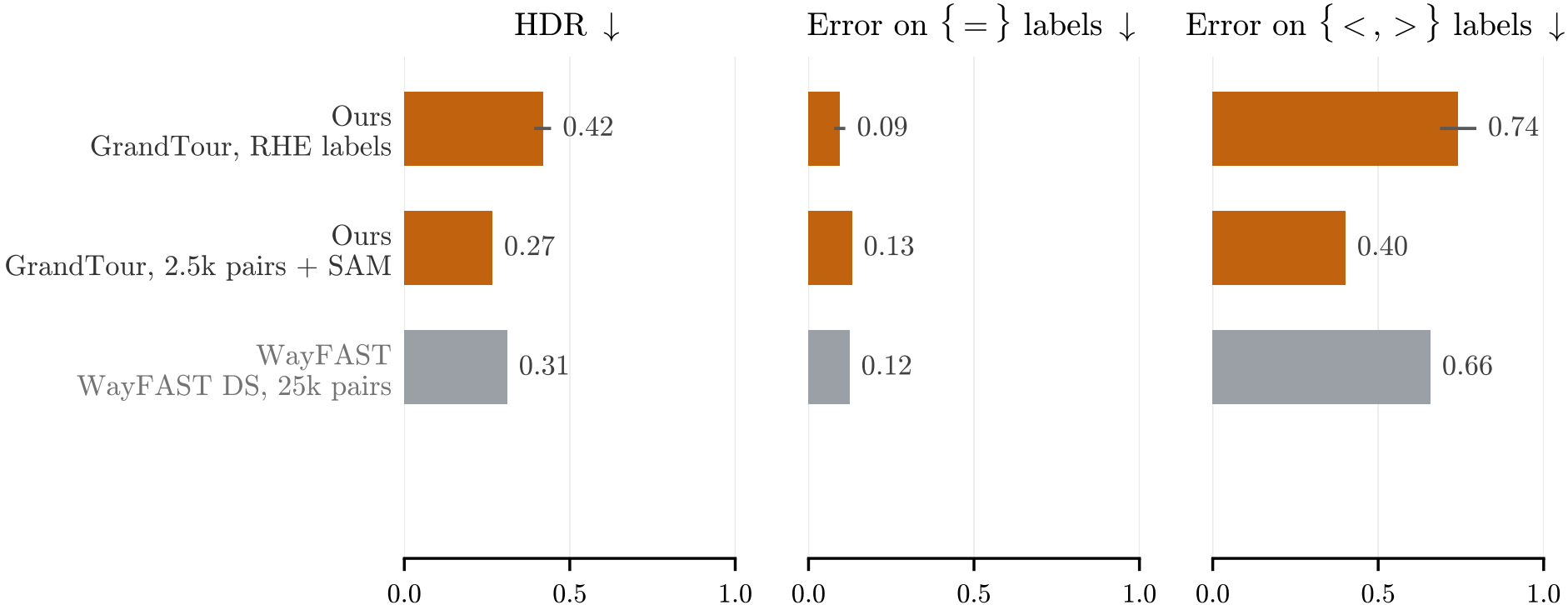
ours: 365 held-out human-labeled pairs, published: as reported on the WayFAST dataset; $\tau = 0.25$

Results: HDR



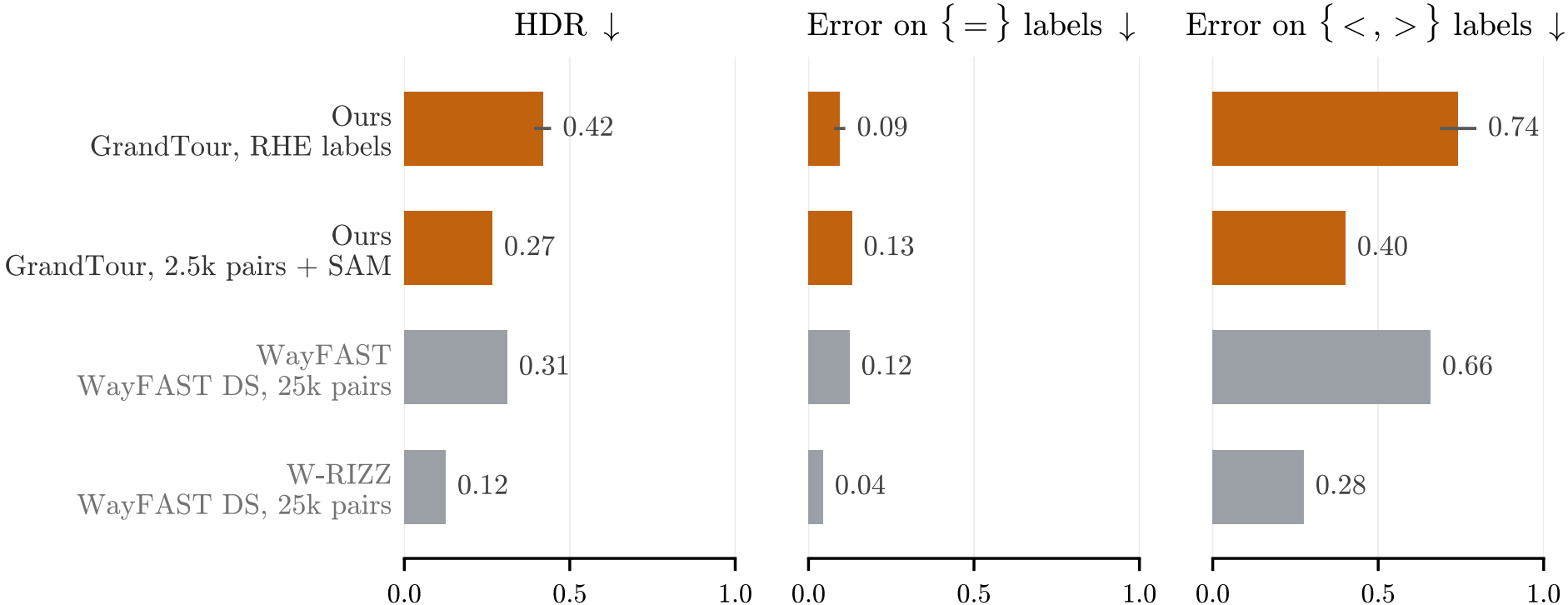
ours: 365 held-out human-labeled pairs, published: as reported on the WayFAST dataset; $\tau = 0.25$

Results: HDR



ours: 365 held-out human-labeled pairs, published: as reported on the WayFAST dataset; $\tau = 0.25$

Results: HDR

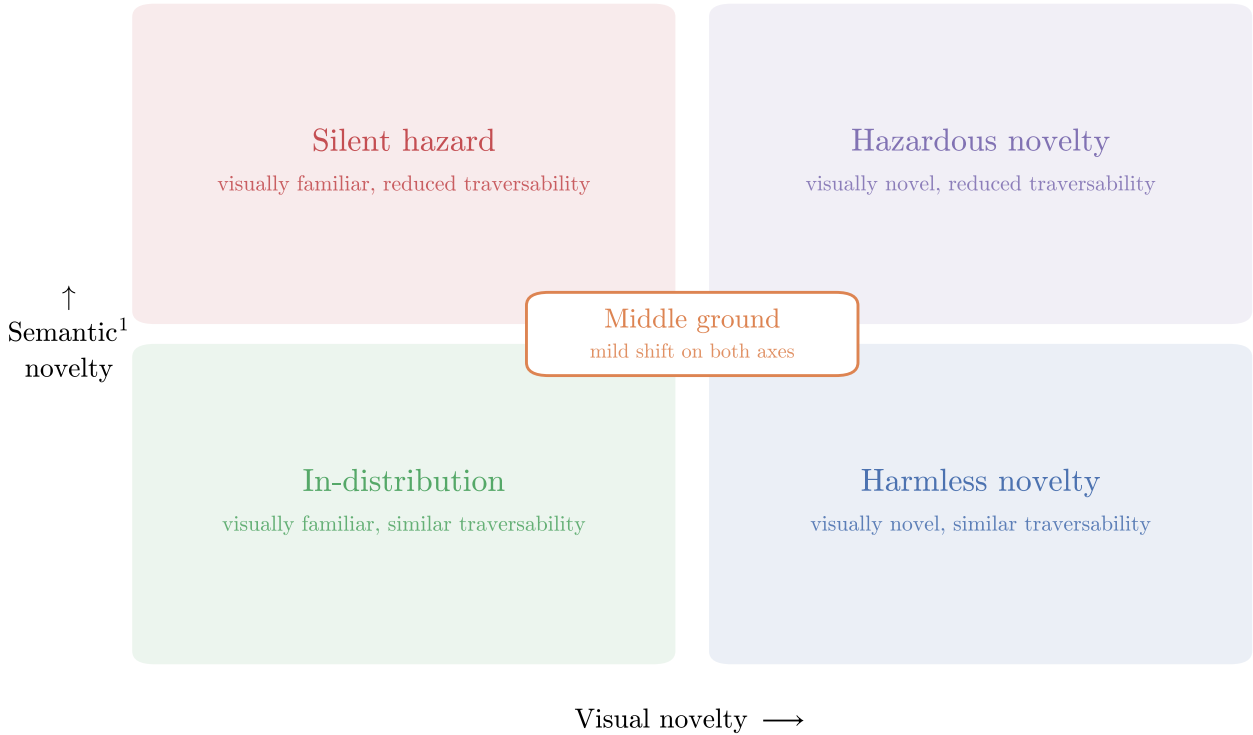


ours: 365 held-out human-labeled pairs, published: as reported on the WayFAST dataset; $\tau = 0.25$

RQ2-RQ6: OOD Detection on the Trained Estimator

OOD-Tiers

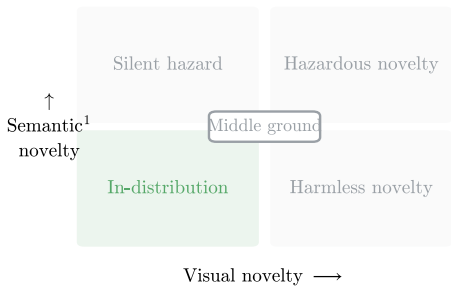
Two Kinds of Novelty



[1] w.r.t. traversability

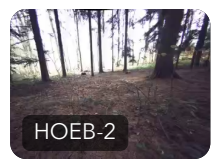
Train and Held-Out

what the network was trained on, and familiar terrain it never trained on

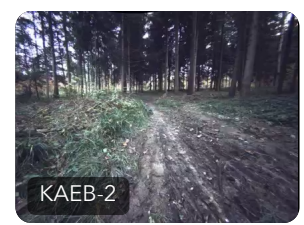


[1] w.r.t. traversability

Train ID

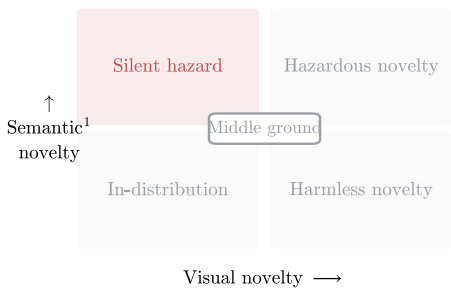


Held-out ID HIL-1, RIV-1: part of the recording



Silent Hazard

looks familiar, harder to walk



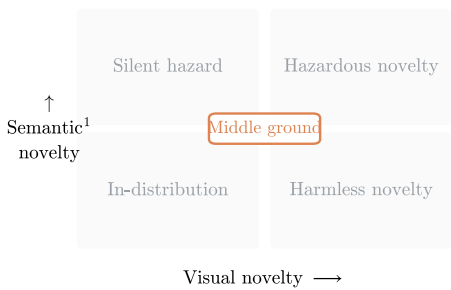
[1] w.r.t. traversability

Near-Roots

Near-Undergrowth KAEB-1: part of the recording

Middle Ground

somewhat new, somewhat harder

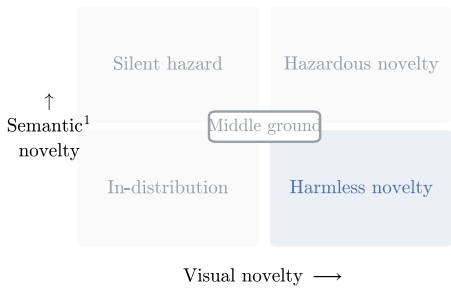


[1] w.r.t. traversability

Medium-Alpine EIG-1, EIG-2, PIL-2: part of the recording

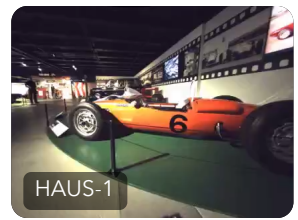
Harmless Novelty

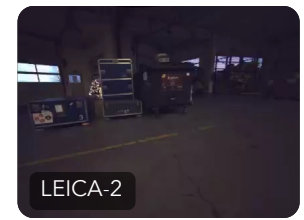
looks new, just as easy



[1] w.r.t. traversability

Far-Indoor LEICA-2: part of the recording

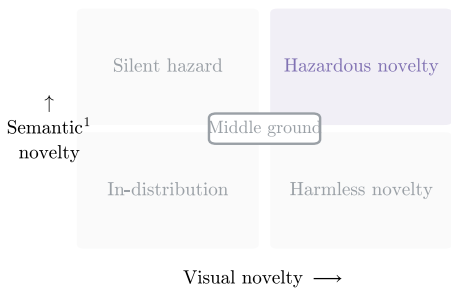




Far-Urban

Hazardous Novelty

looks new and harder



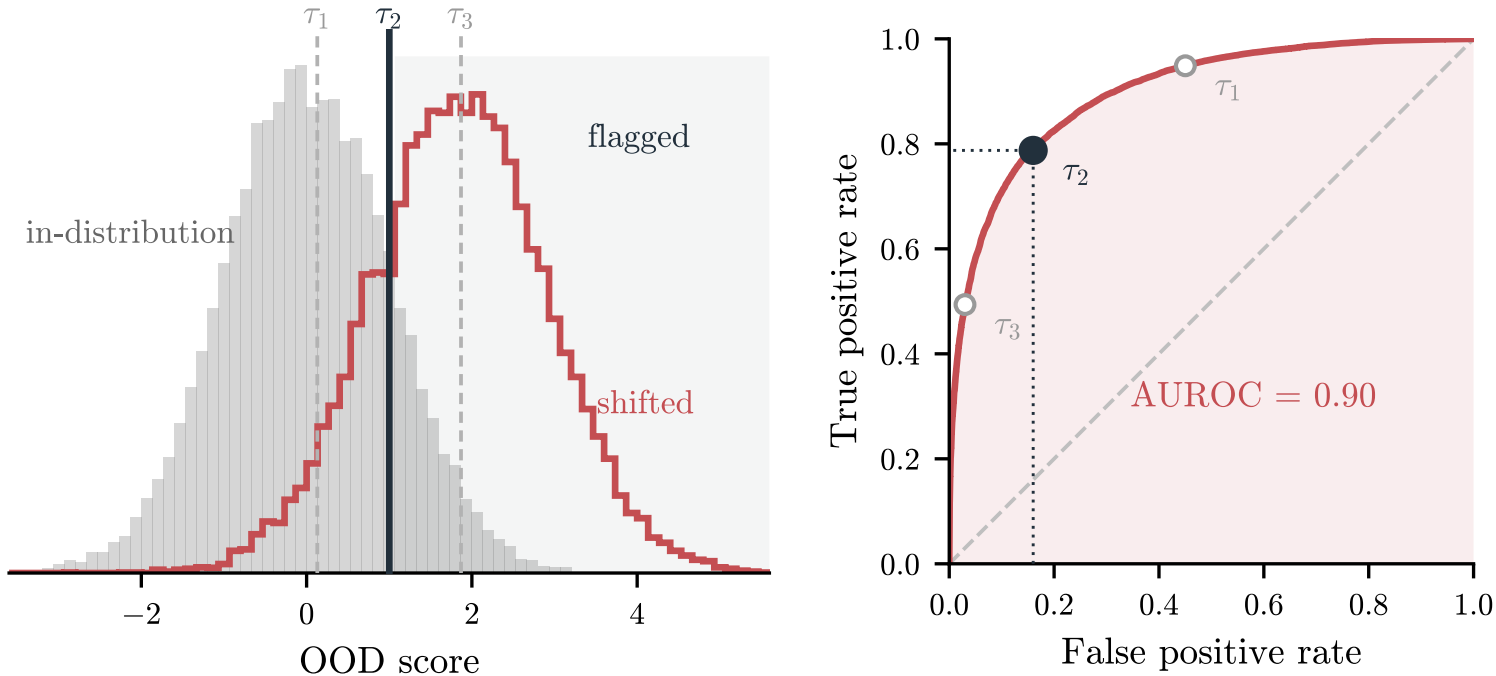
[1] w.r.t. traversability

Far-Snow

Far-Construction CON-3, HEAP-1: part of the recording

Metrics

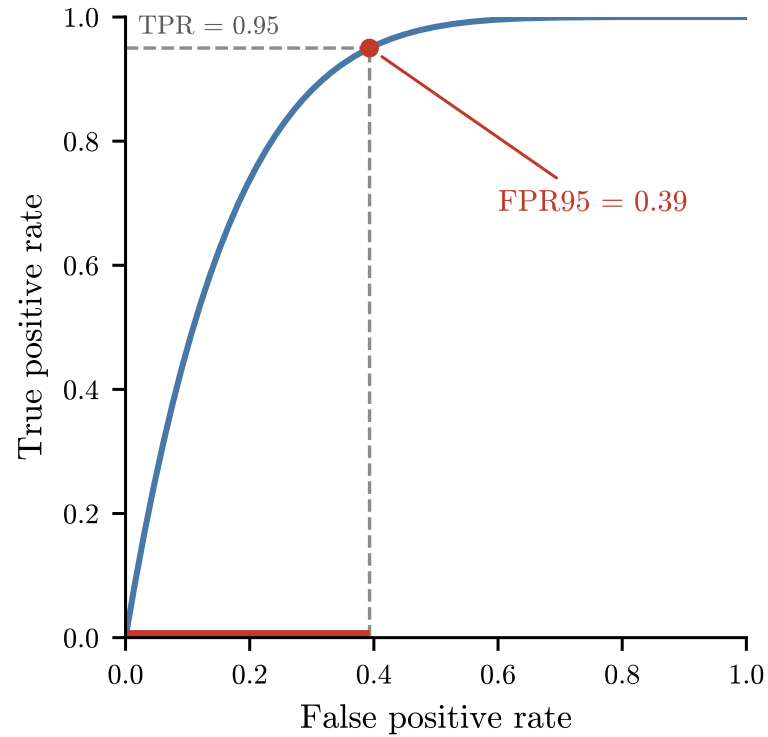
AUROC (Area Under the Receiver Operating Characteristic Curve)



AUROC is the area under the whole curve, so it needs no threshold.

FPR95 (False Positive Rate at True Positive Rate of 95%)

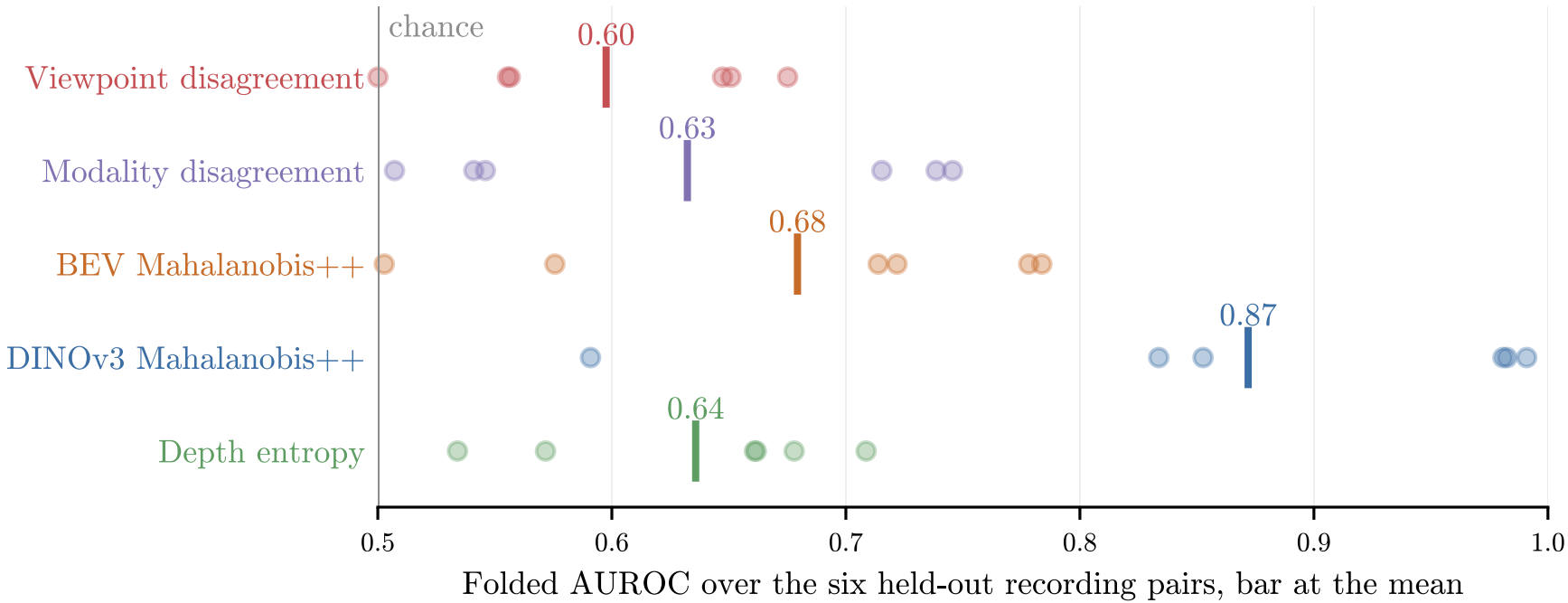
- Set the threshhold so that the detector catches 95% of the OOD data.
- How many false alarms?



How the Session Floor Is Computed

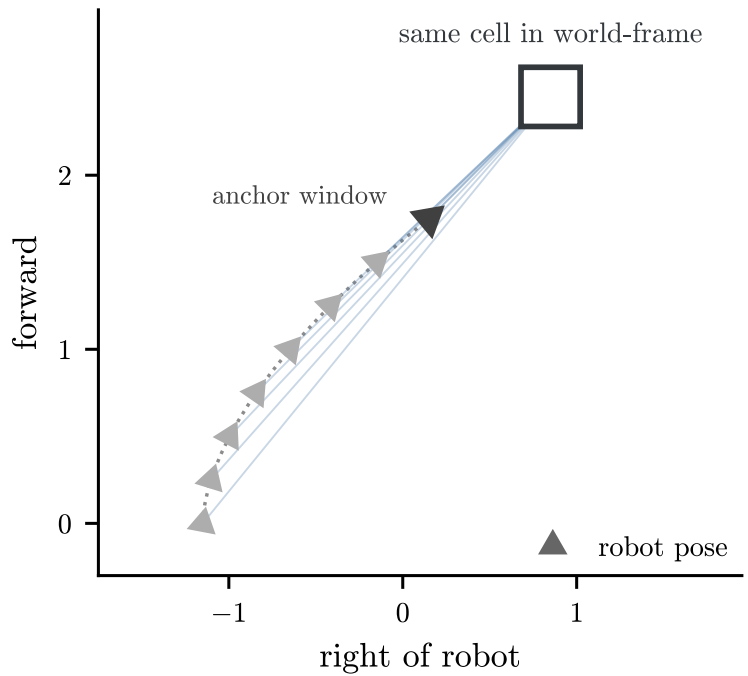
- 4 held-out ID recordings: ALB-3, HIL-1, KAEB-2, RIV-1
- Take every (unordered) pair of them (6 pairs)
- For each detector: AUROC on each pair, average the 6 AUROCs
- Answers: Does the detector separate ID recordings from each other as well?

Floor of Every Detector

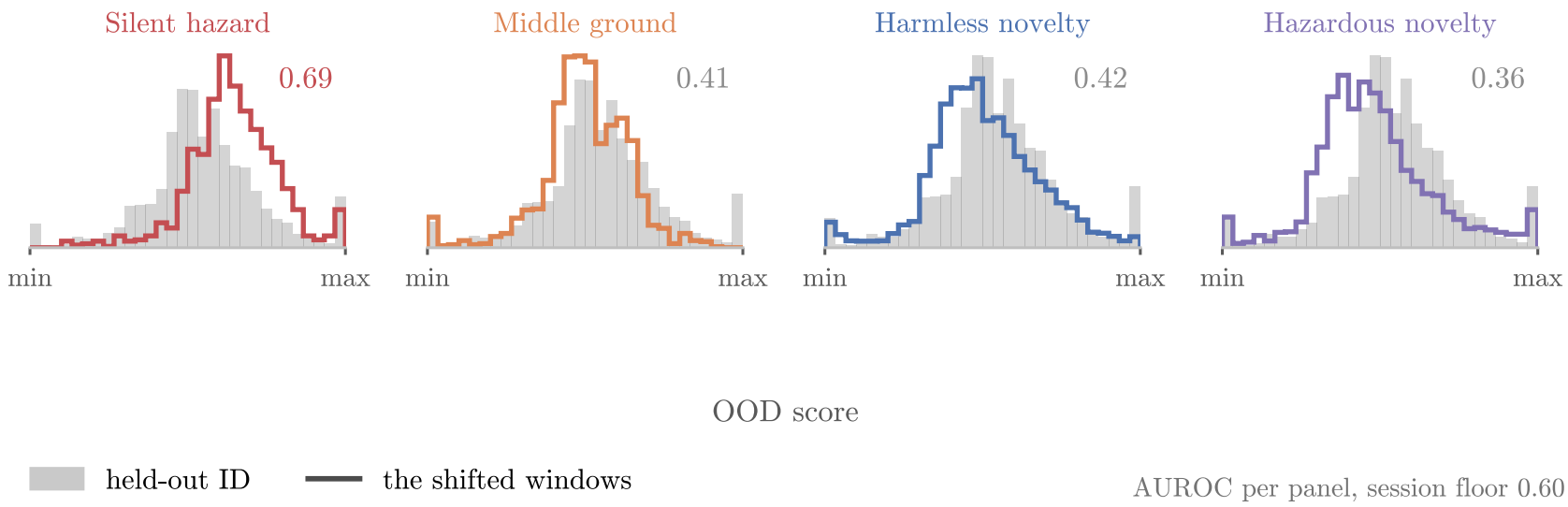


Detectors

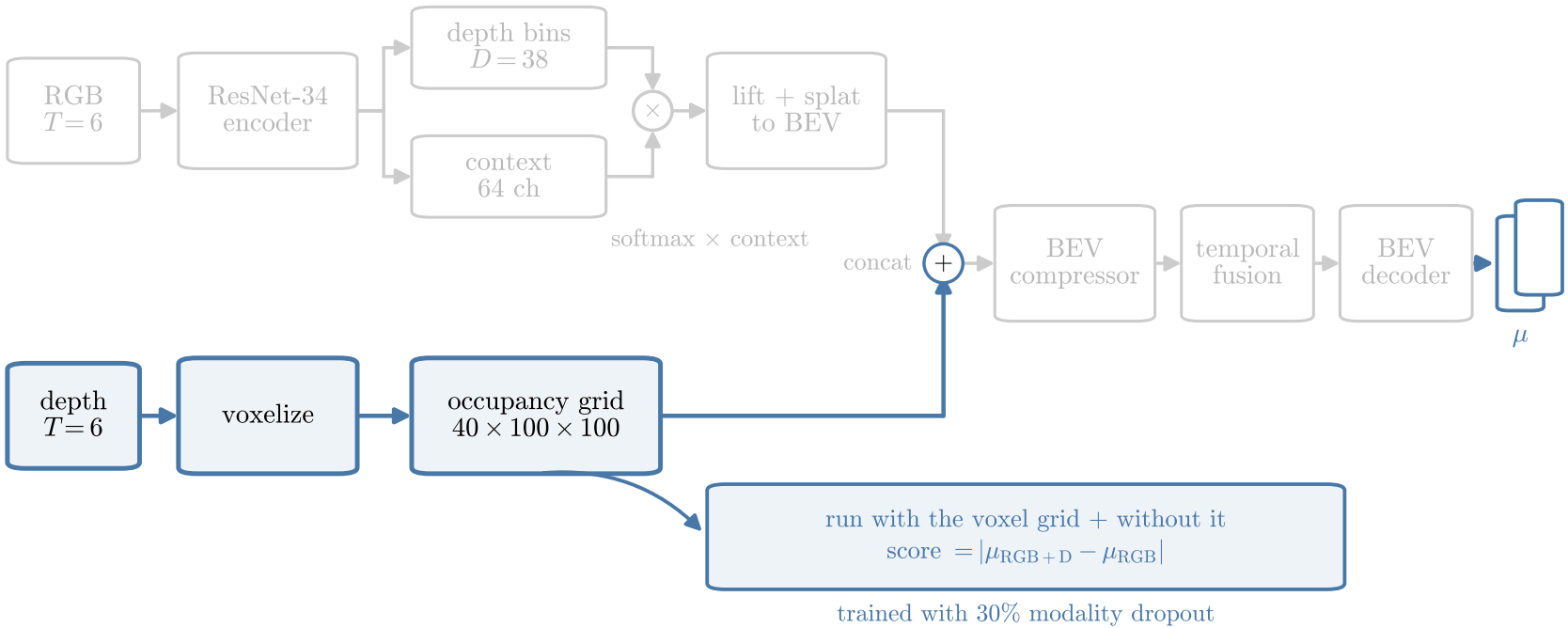
Viewpoint Disagreement



Viewpoint Disagreement - Scores

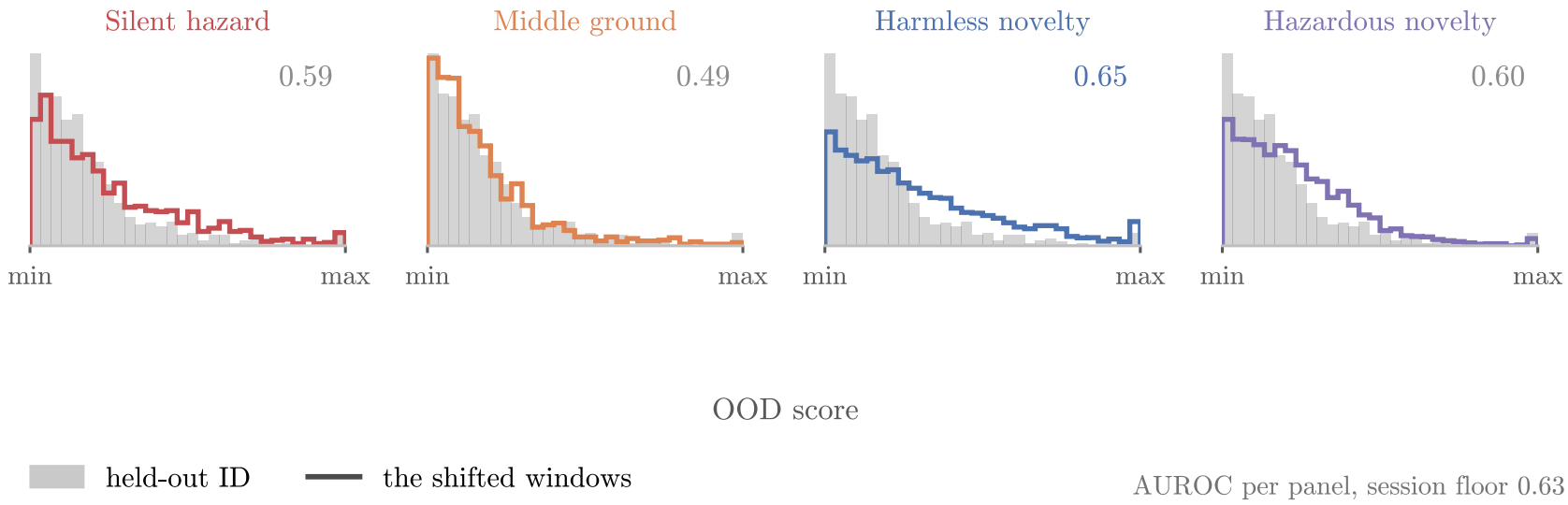


Modality Disagreement



Similar to ensemble approaches.

Modality Disagreement - Scores

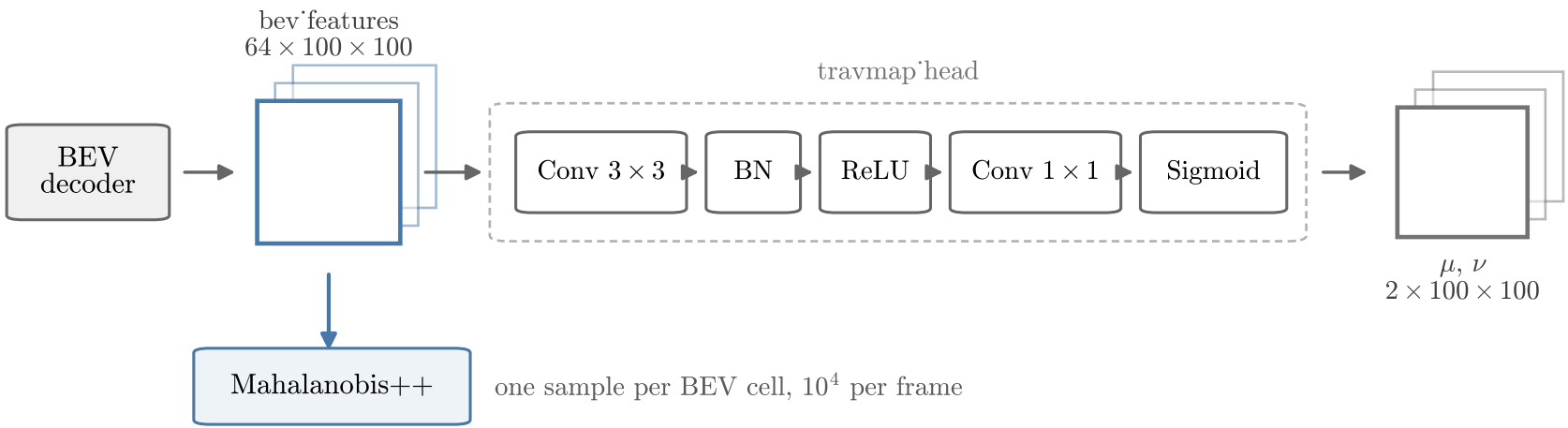


ARXIV:2505.18032 · 2025

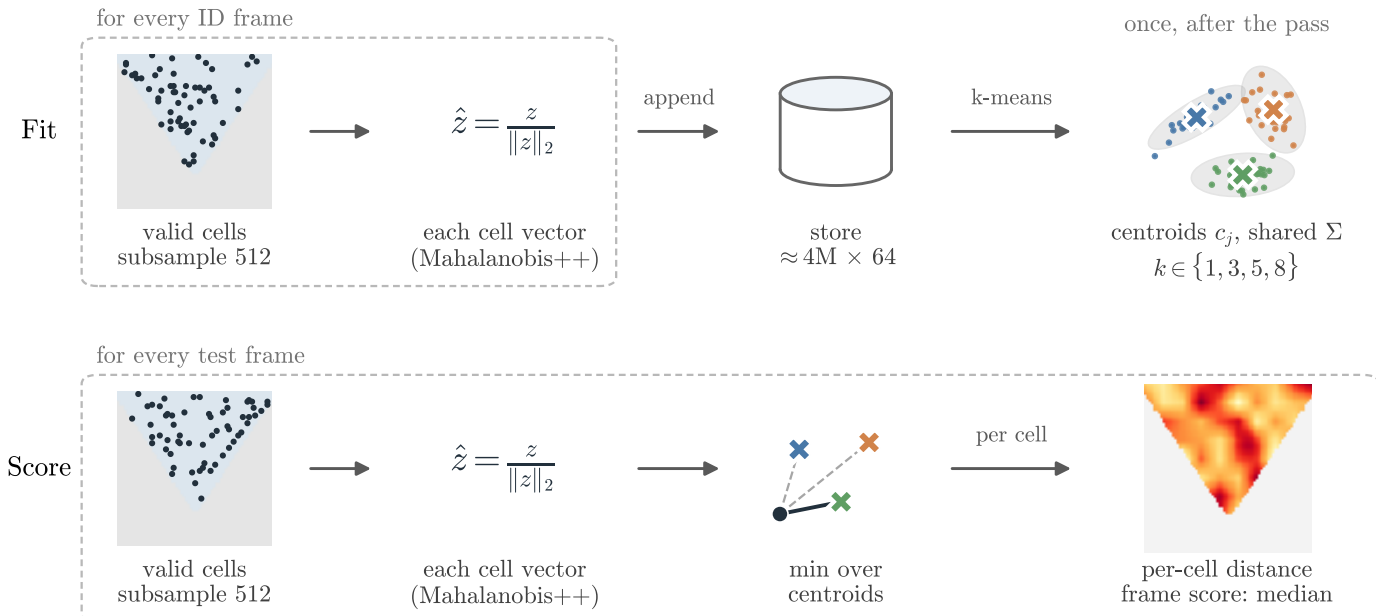
Mahalanobis++: Improving OOD Detection via Feature Normalization

Maximilian Mueller, Matthias Hein

WayFASTER Features



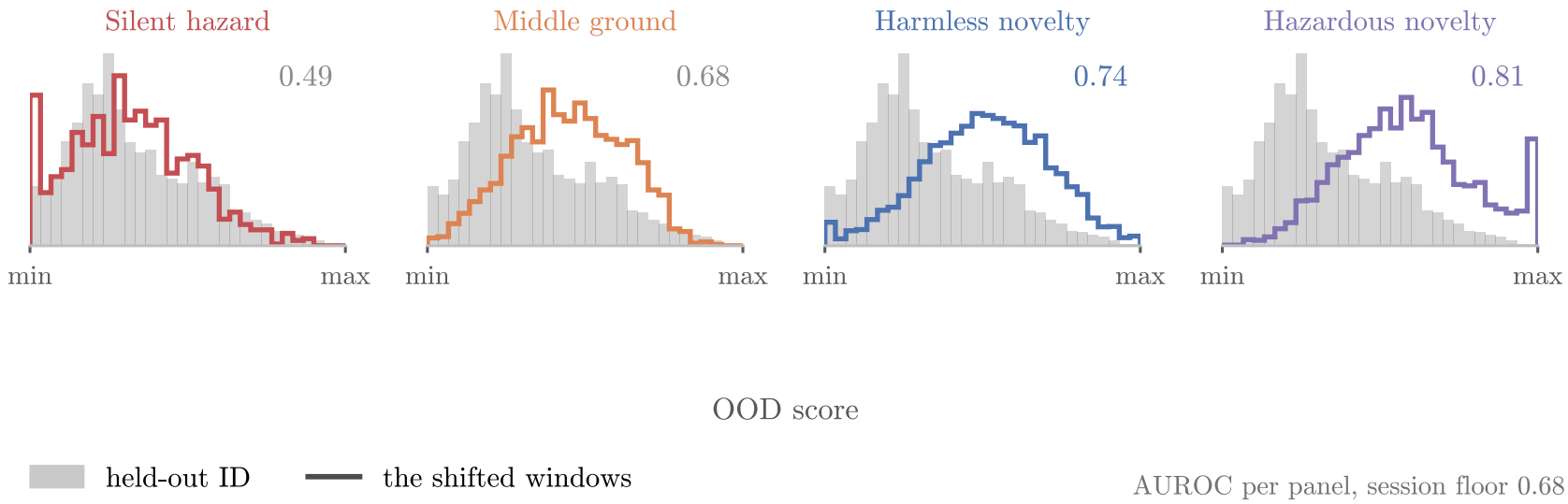
Mahalanobis++



In classification problems, the label classes is used for clustering. This is not applicable to our regression problem.

Mueller and Hein, 2025. The distance itself is on an extra slide.

BEV Mahalanobis++ - Scores



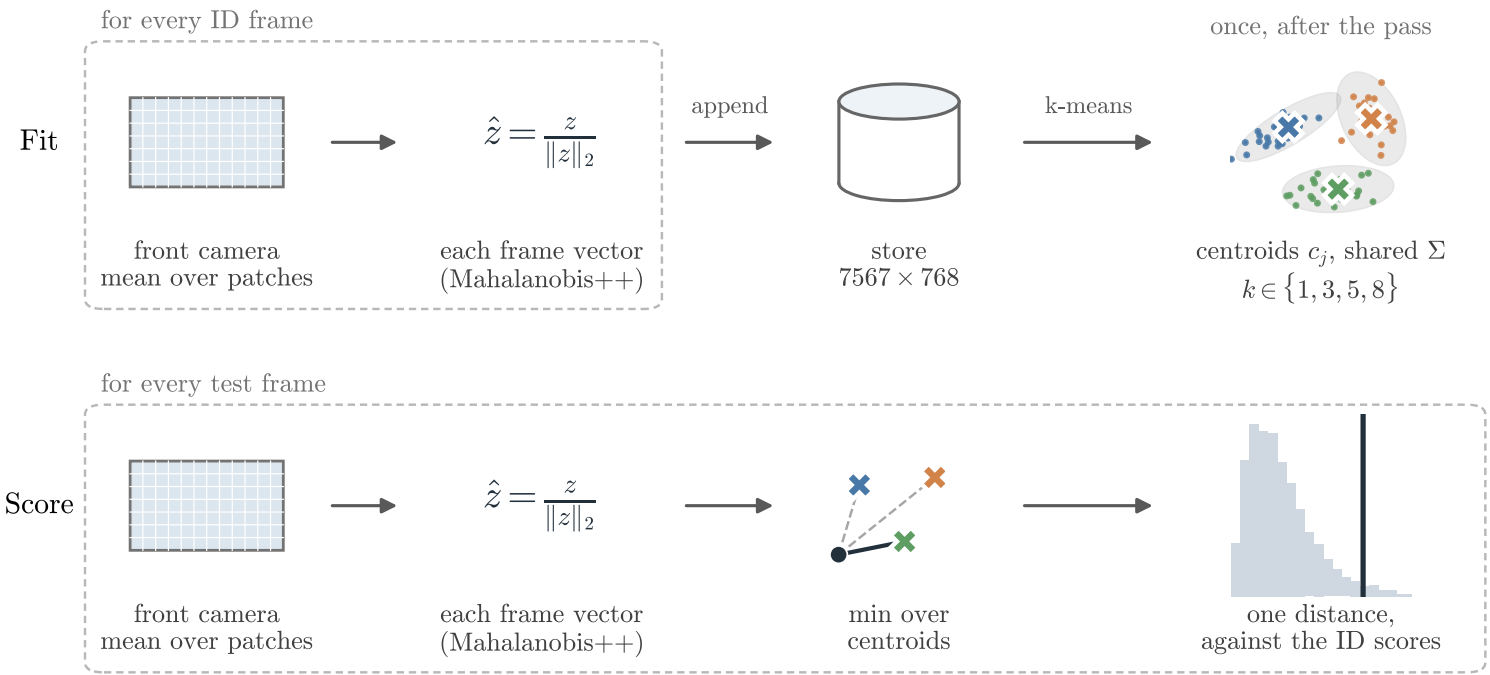
ARXIV:2508.10104 · 2025

DINOv3

Siméoni, Vo, Seitzer, Baldassarre, Oquab et al.

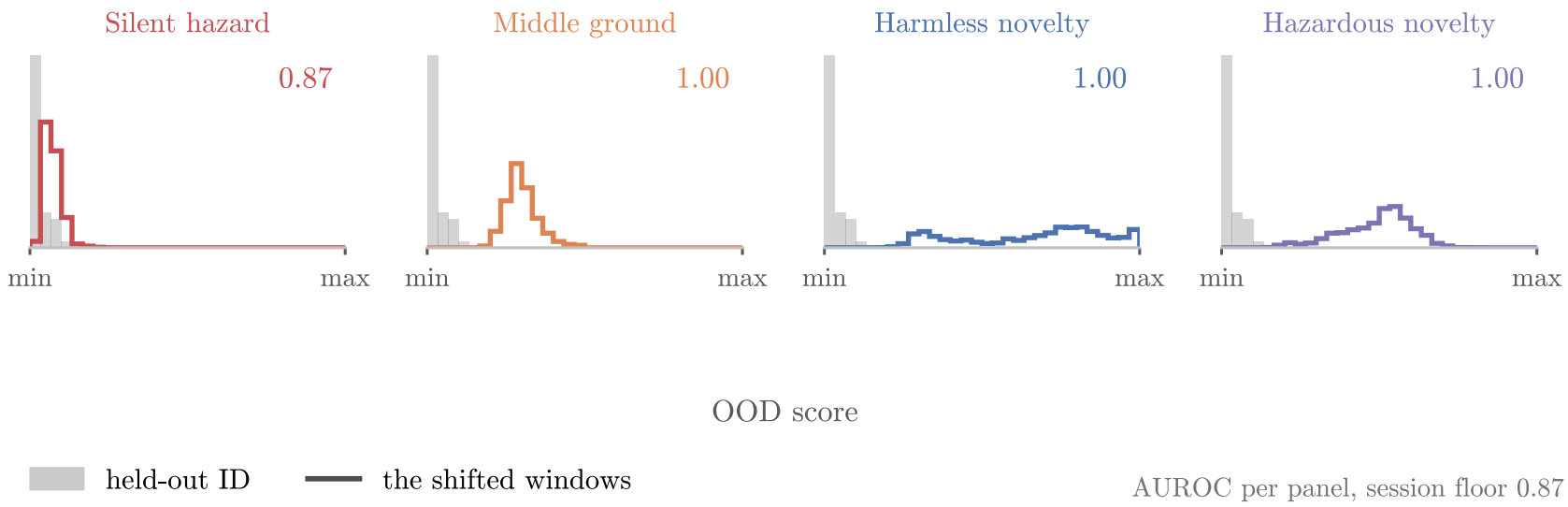
Self-supervised image feature extractor

The Same, on DINOv3 Features

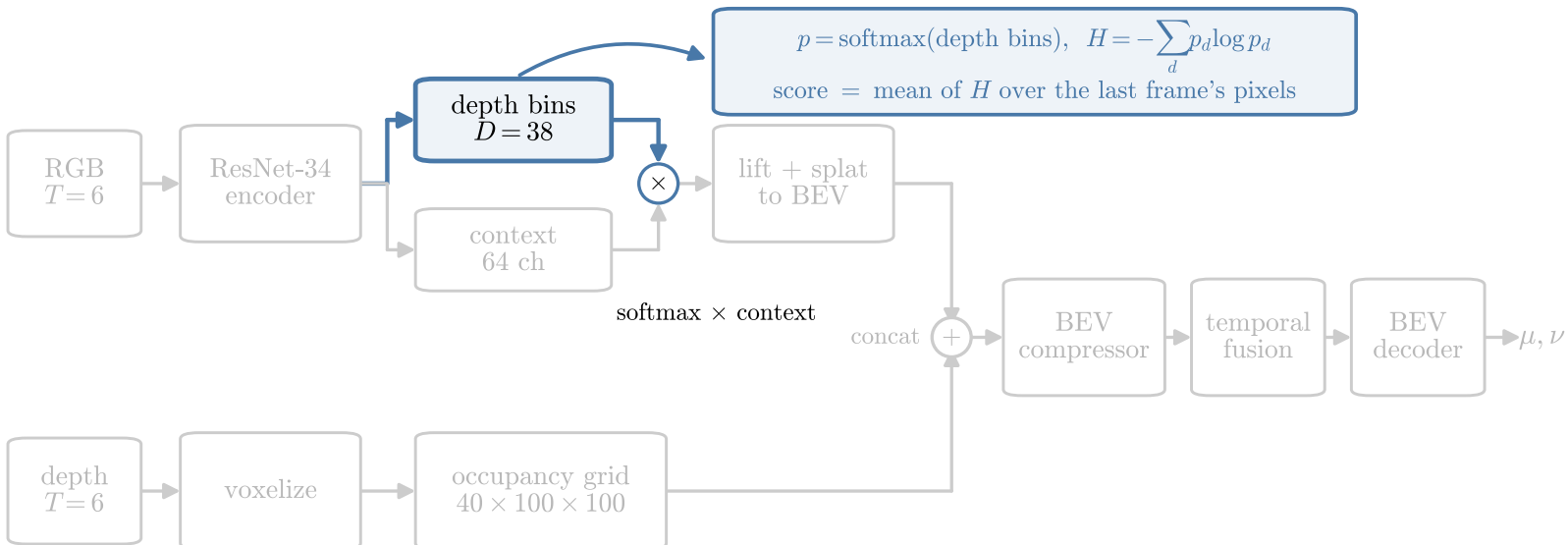


DINOv3: Siméoni et al., 2025.

DINOv3 Mahalanobis++ - Scores

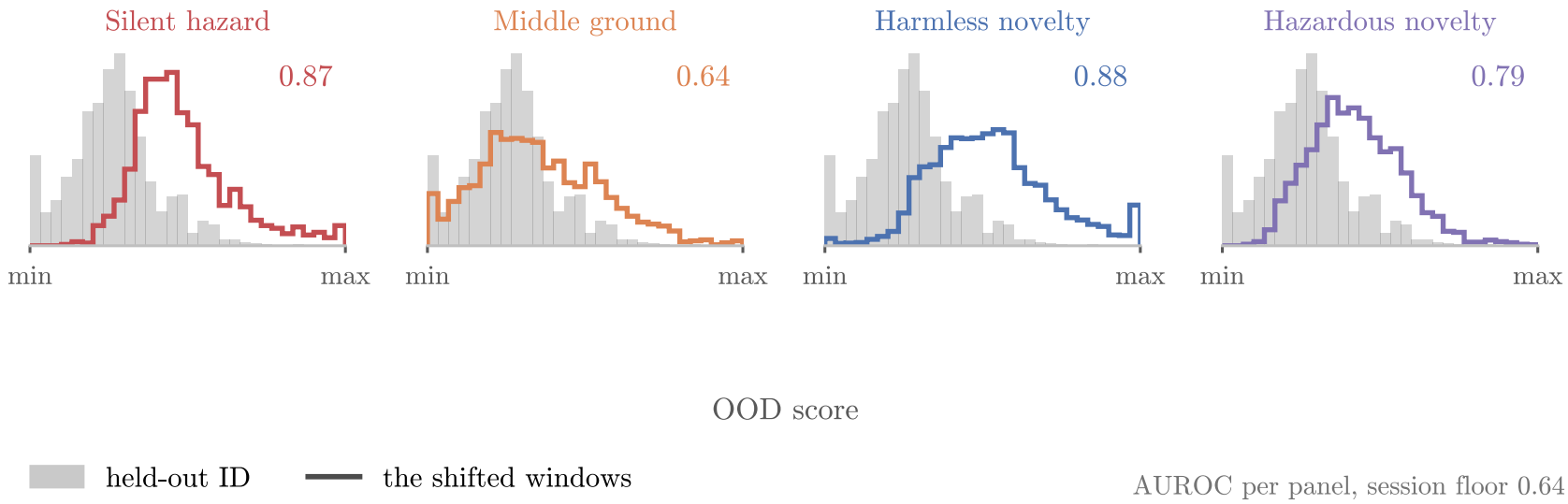


Depth Entropy



Hypothethis: Unfamiliar terrain --> Higher depth entropy.

Depth Entropy - Scores



All Detectors: AUROC and FPR95

	AUROC (higher is better)					FPR95 (lower is better)			
	Silent hazard	Middle ground	Harmless novelty	Hazardous novelty	session floor	Silent hazard	Middle ground	Harmless novelty	Hazardous novelty
Viewpoint disagreement	0.69	0.41	0.42	0.36	0.60	0.83	0.95	0.95	0.96
Modality disagreement	0.59	0.49	0.65	0.60	0.63	0.93	0.95	0.92	0.92
BEV Mahalanobis++	0.49	0.68	0.74	0.81	0.68	0.98	0.81	0.76	0.55
DINOv3 Mahalanobis++	0.87	1.00	1.00	1.00	0.87	0.23	0.00	0.00	0.00
Depth entropy	0.87	0.64	0.88	0.79	0.64	0.44	0.92	0.49	0.68

Conclusion

Limitations

- Limited Dataset size (1 h 40 min)
- Relative label pairs cover ID missions only, from a single annotator
- Missions were manually sorted into quadrants along the traversability axis
- No test set and no evaluation on an actual robot

Future Work

Future Work: Establish a Shared Benchmark for Traversability Estimation

Rationale:

- Identifying a good OOD detector is near-impossible without a shared benchmark
- Only evaluating on a robot is too imprecise

This would require:

- Selection of appropriate Datasets
- Selection of label definitions

ARXIV:2606.19675 · 2026

ForEnt: A Multi-Modal Dataset for Characterizing Quadruped Robot Entrapments in Forest Environments

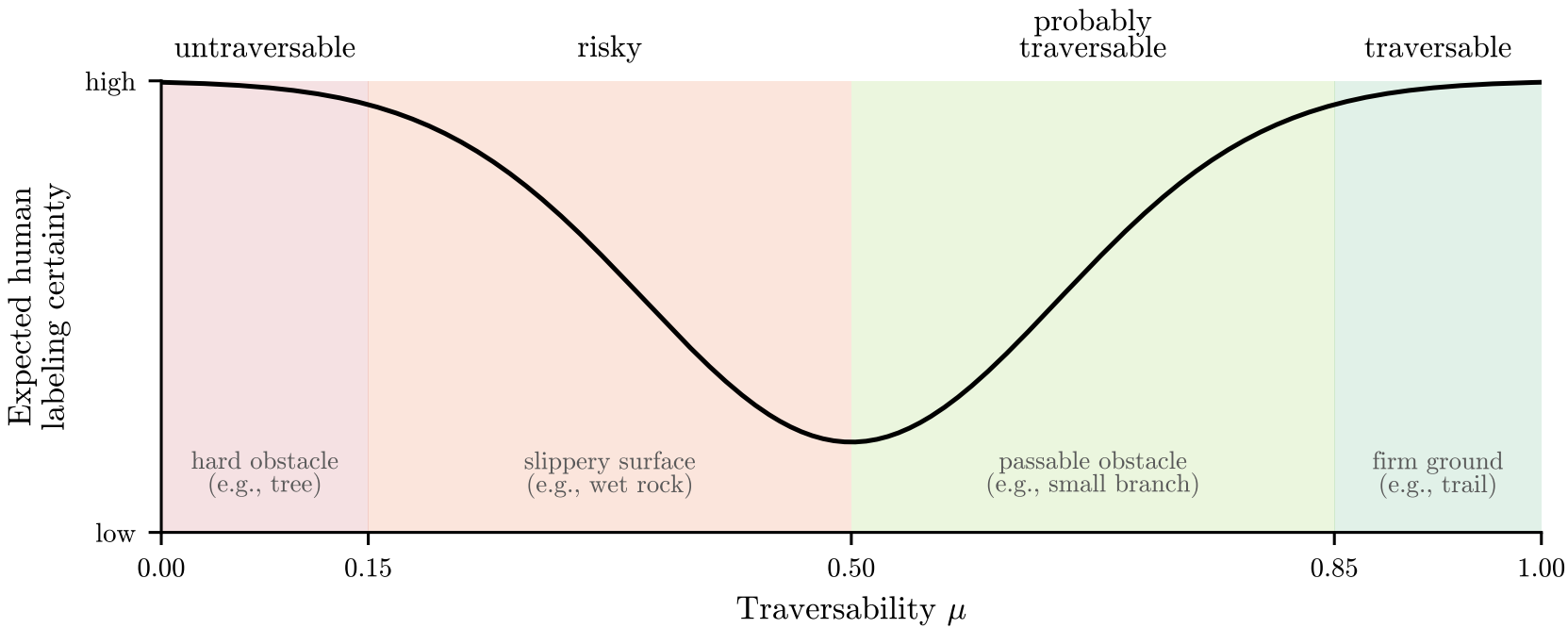
Natapat Kirdwichai, Danesh Tarapore

Future Work: Explore Combining Relative Human Labels with Proprioceptive Labels

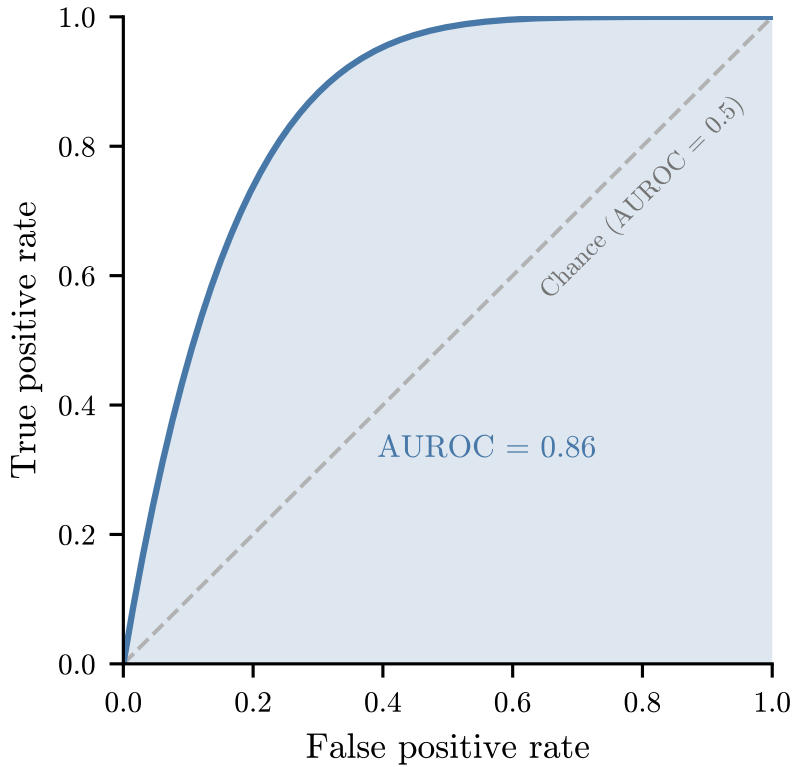
Thank you!

Extra Slides

Limits of Proprioceptive Labels

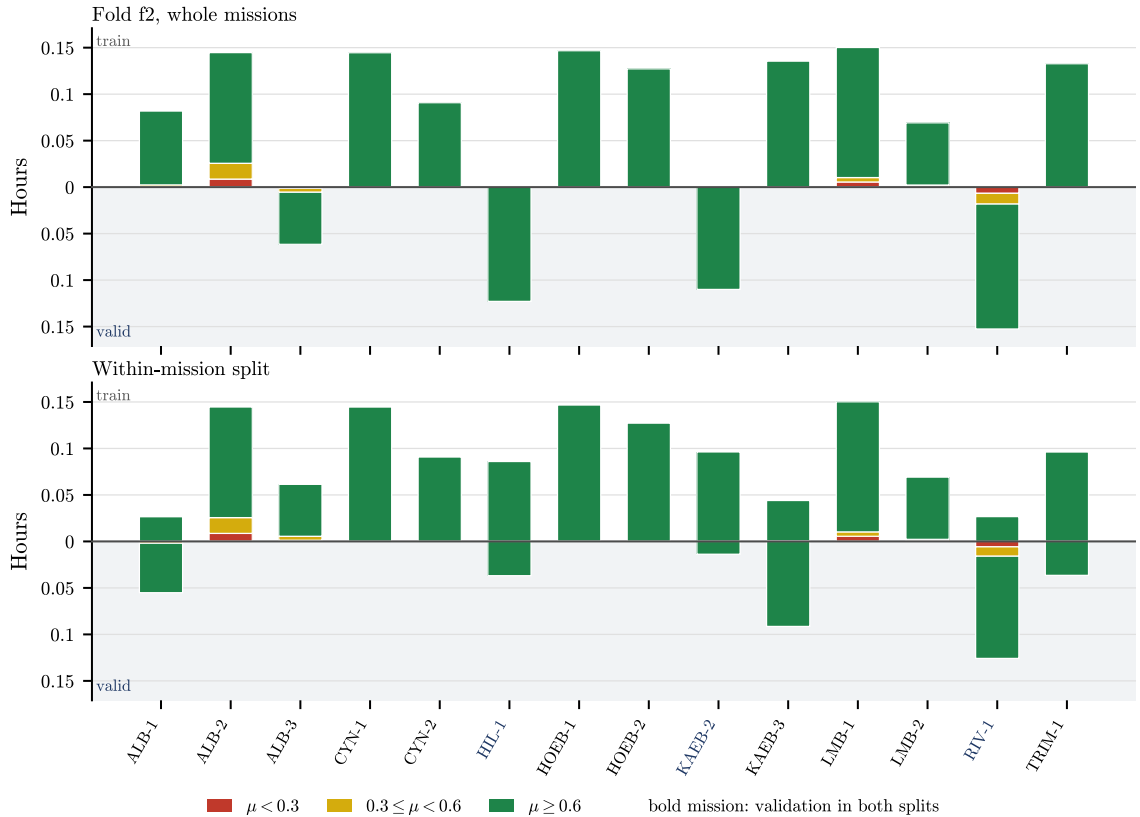


Evaluation Protocol

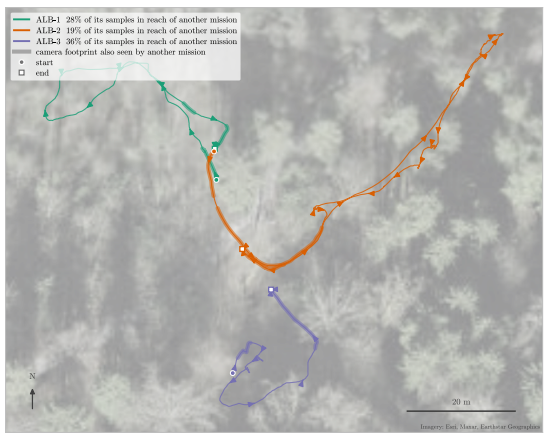


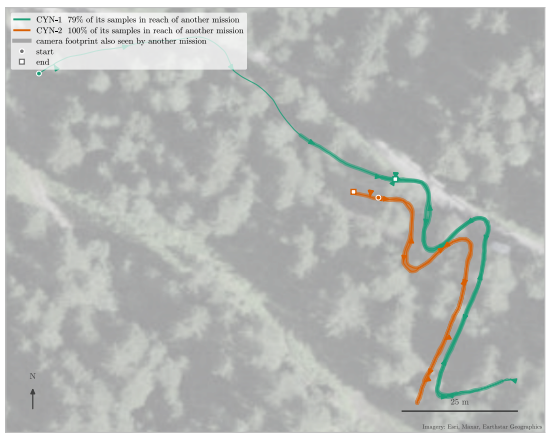
- Treat as binary classification
- **AUROC, risky ($\mu < 0.5$) against the rest.** Threshold-free, measures ordering quality.

Train/Validation Split Selection

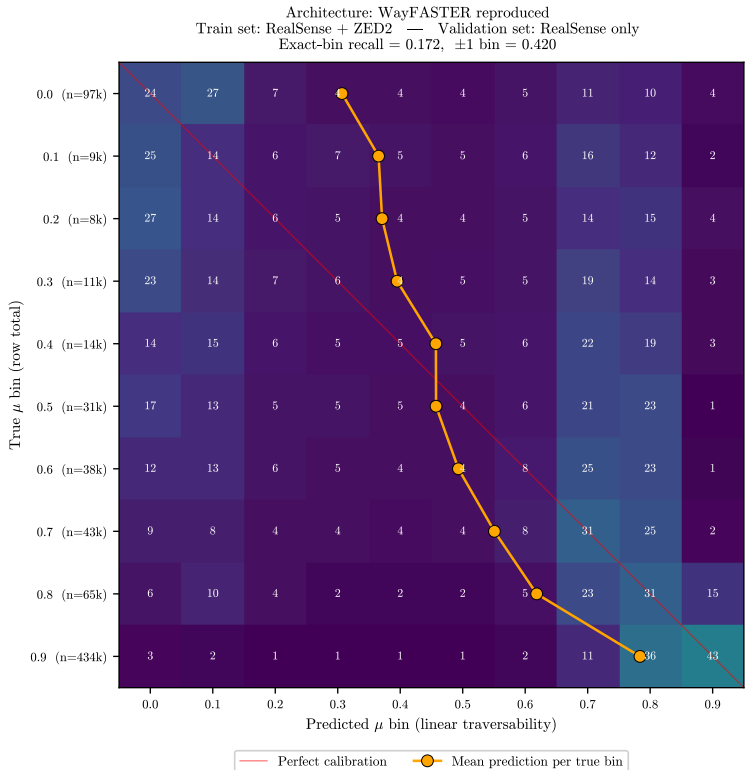


Measuring Ground Overlap

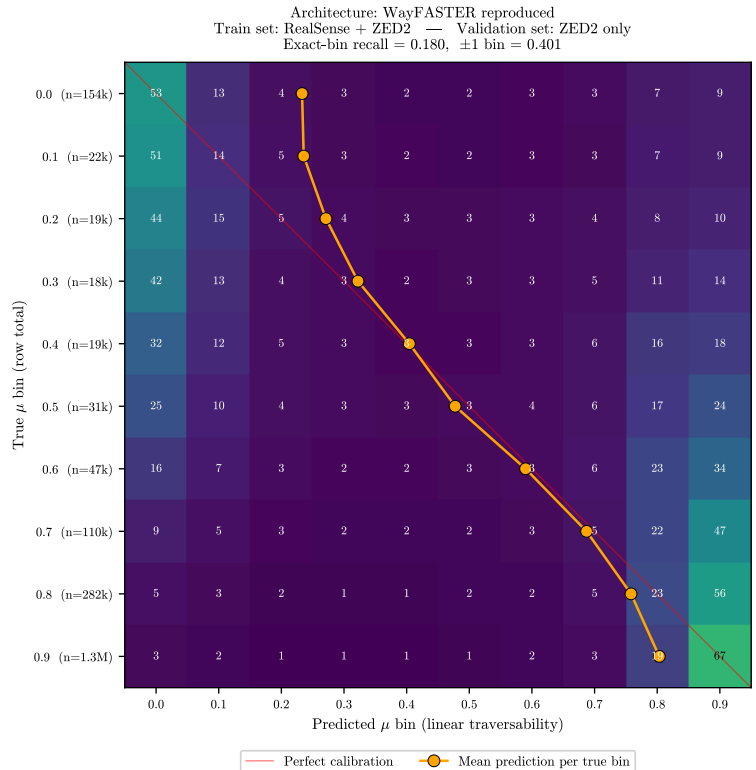




WayFASTER: Choice of Validation Set

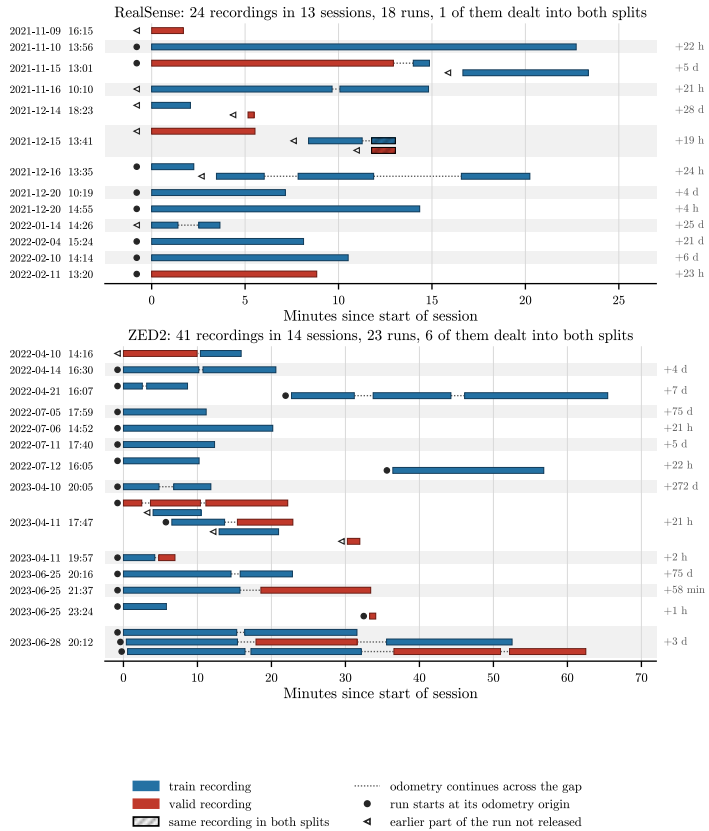


WayFast validation

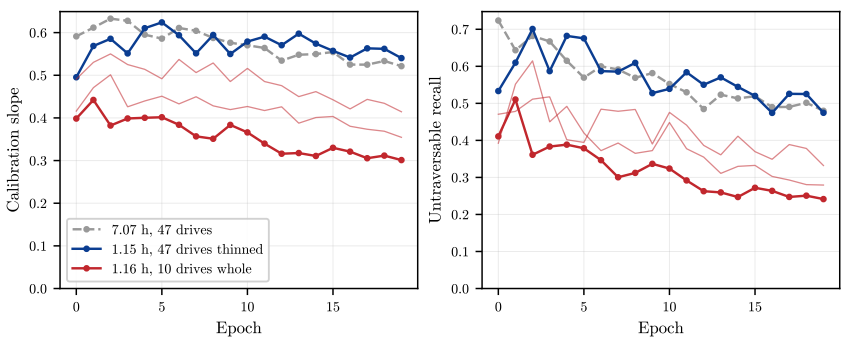


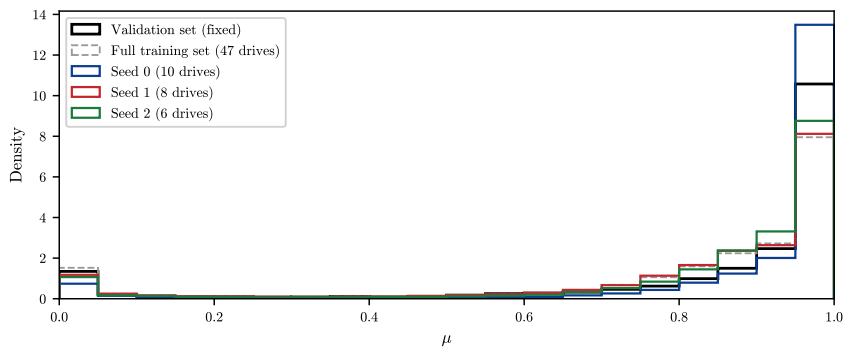
WayFASTER validation

WayFASTER: Recording Timeline

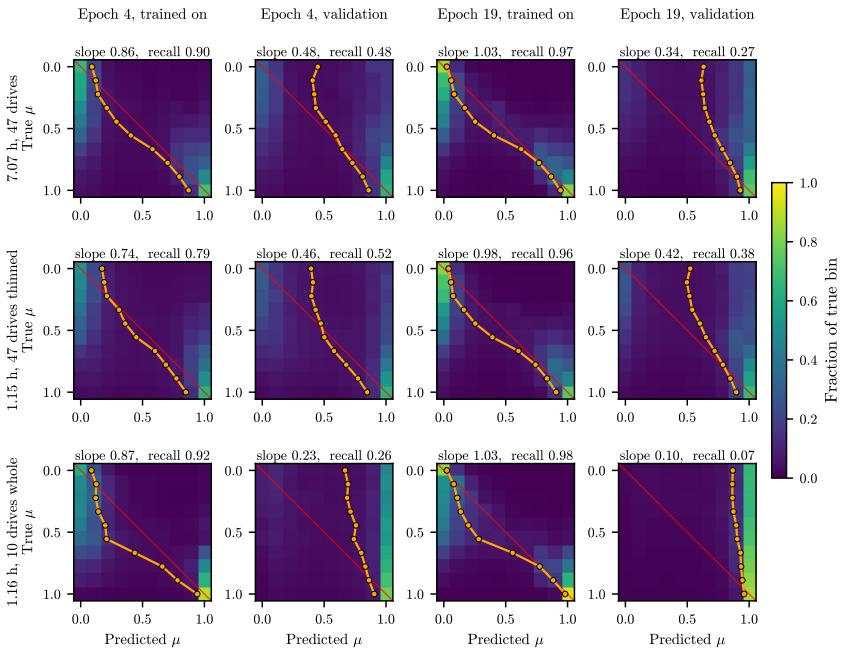


WayFASTER: Data Volume

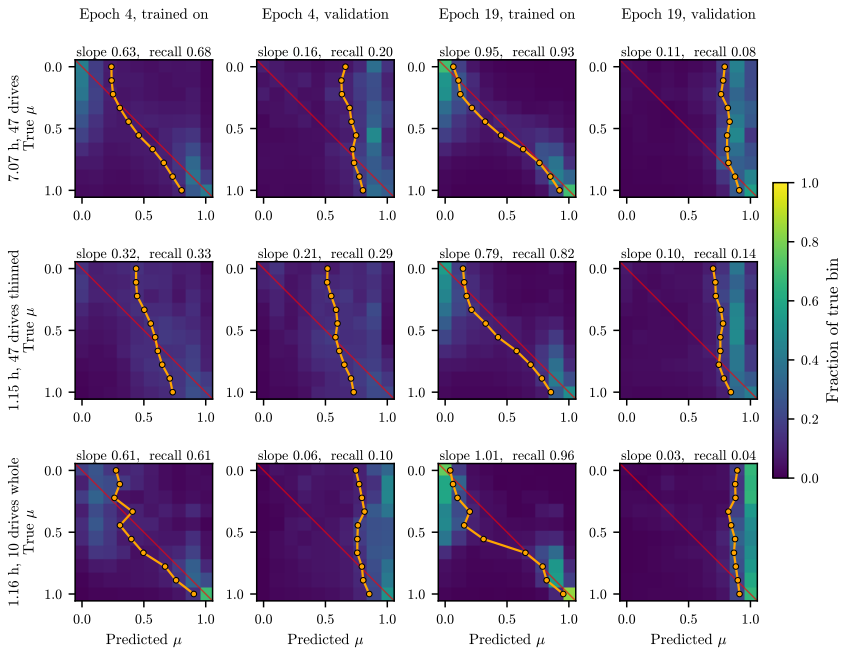




WayFASTER: Memorization

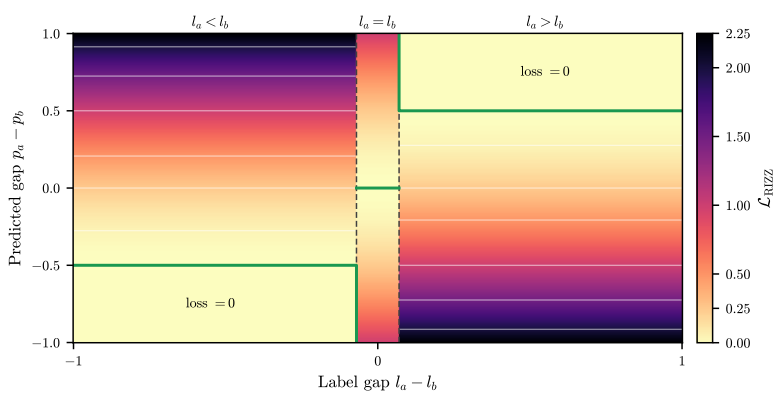


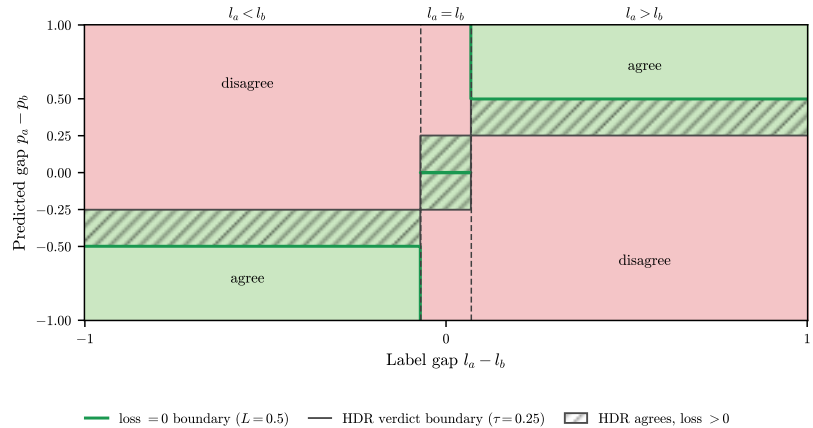
visible cells



blind cells

W-RIZZ: Full Loss Definition

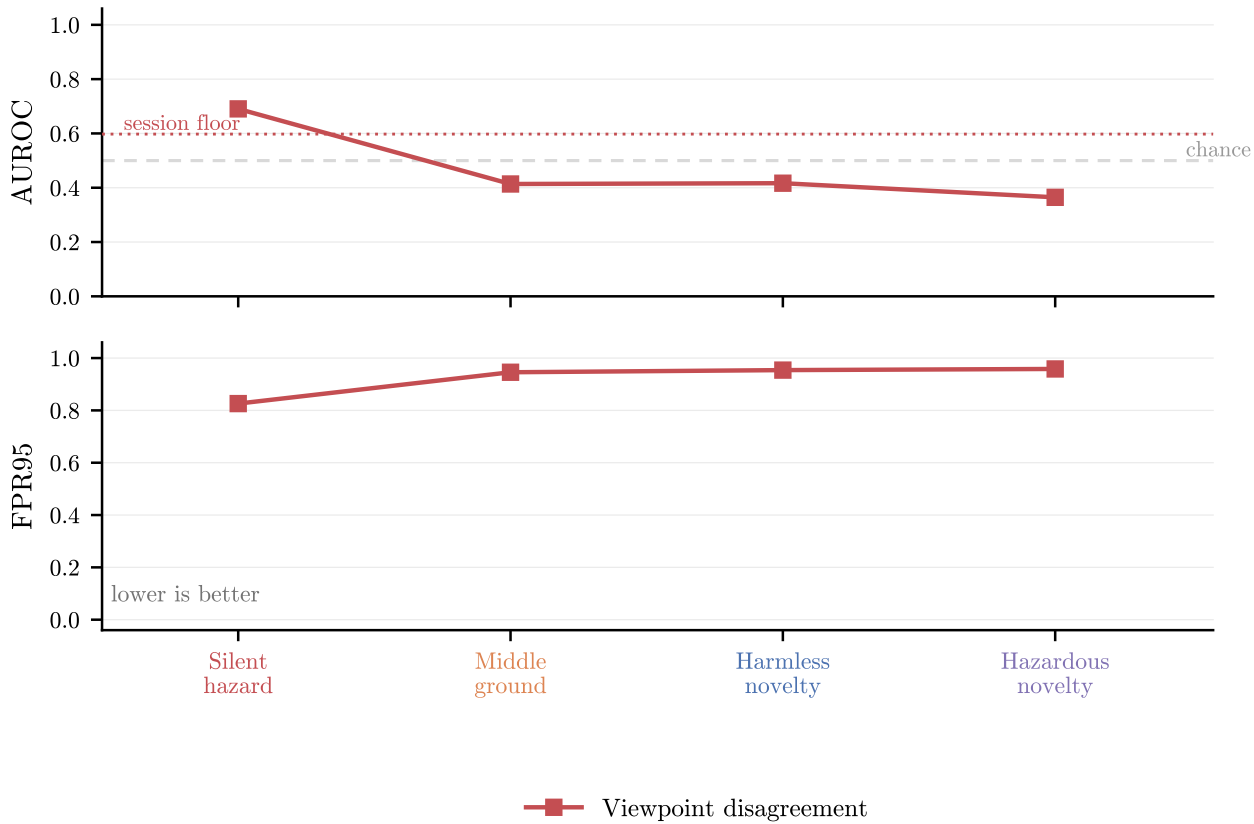




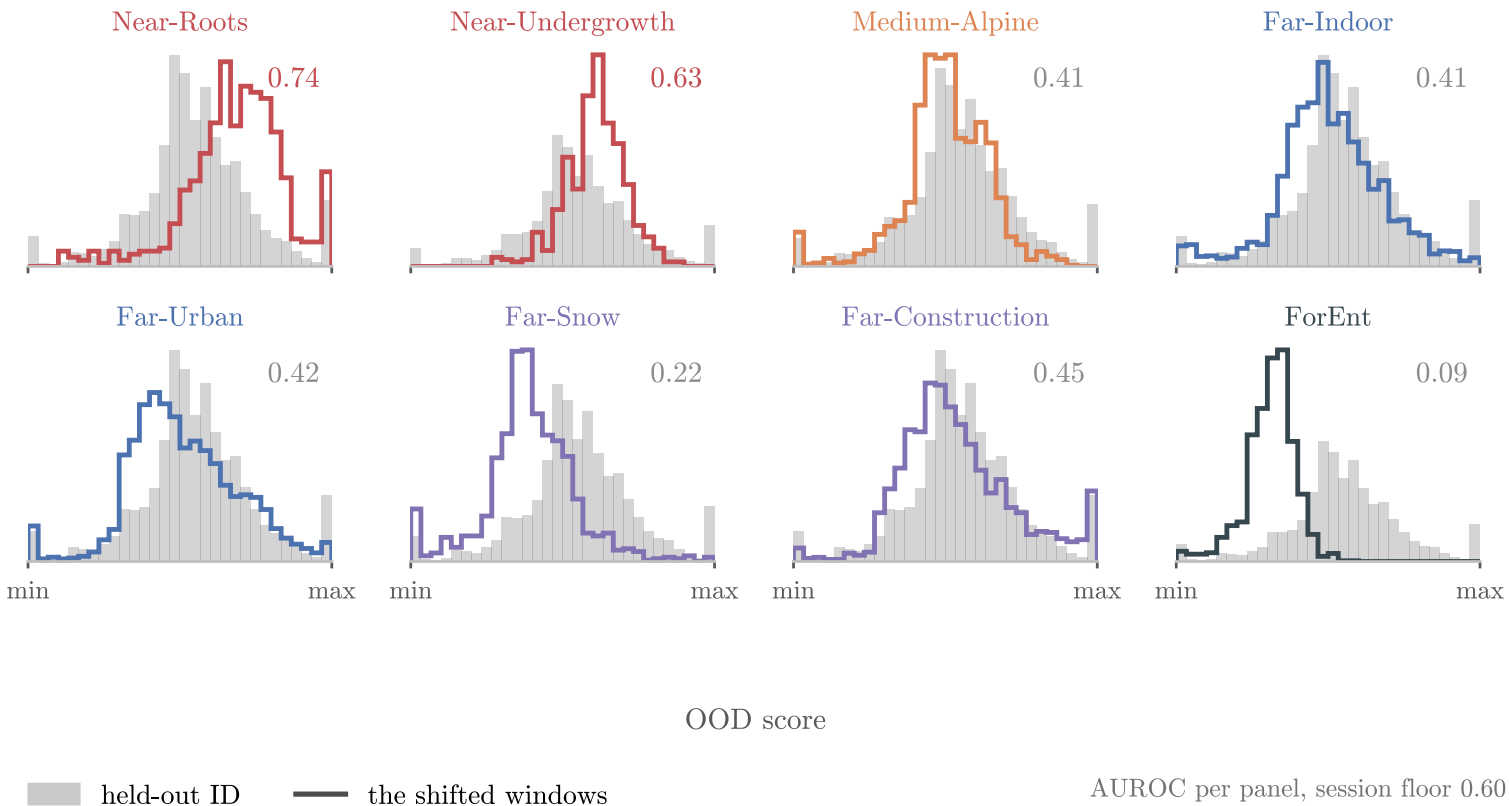
OOD Detectors on ID Data

Each ID recording refit and scored leave-one-out.

Viewpoint Disagreement - AUROC and FPR95



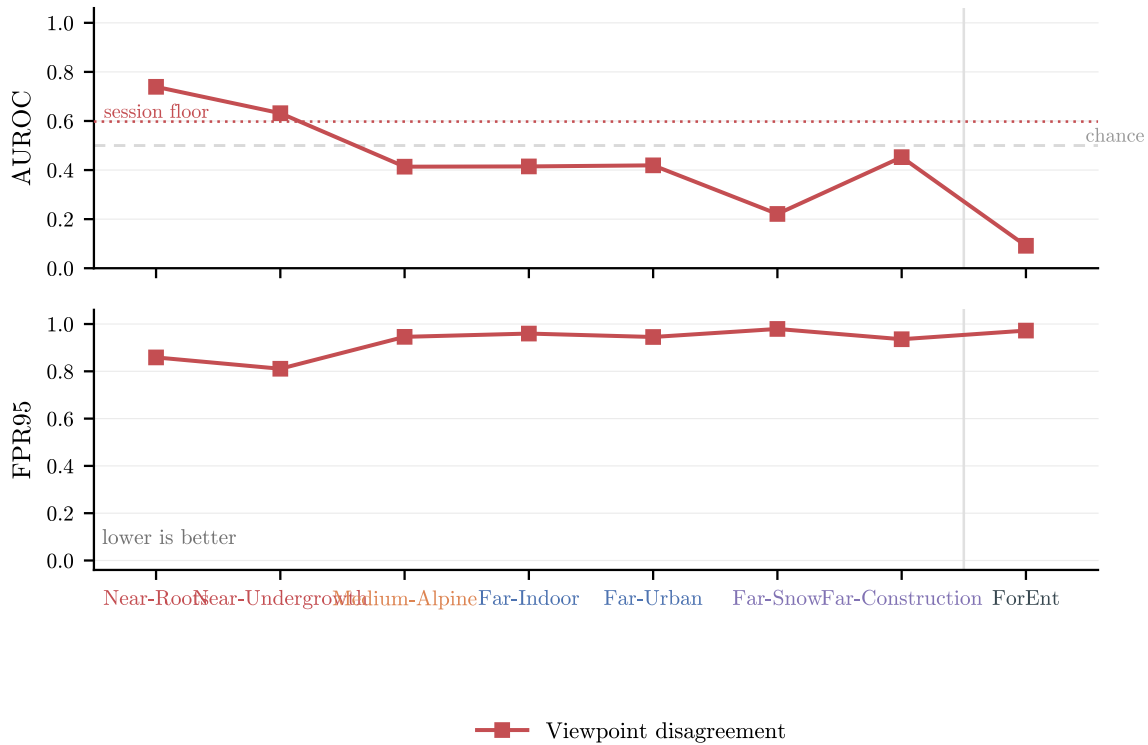
Viewpoint Disagreement - Every Tier



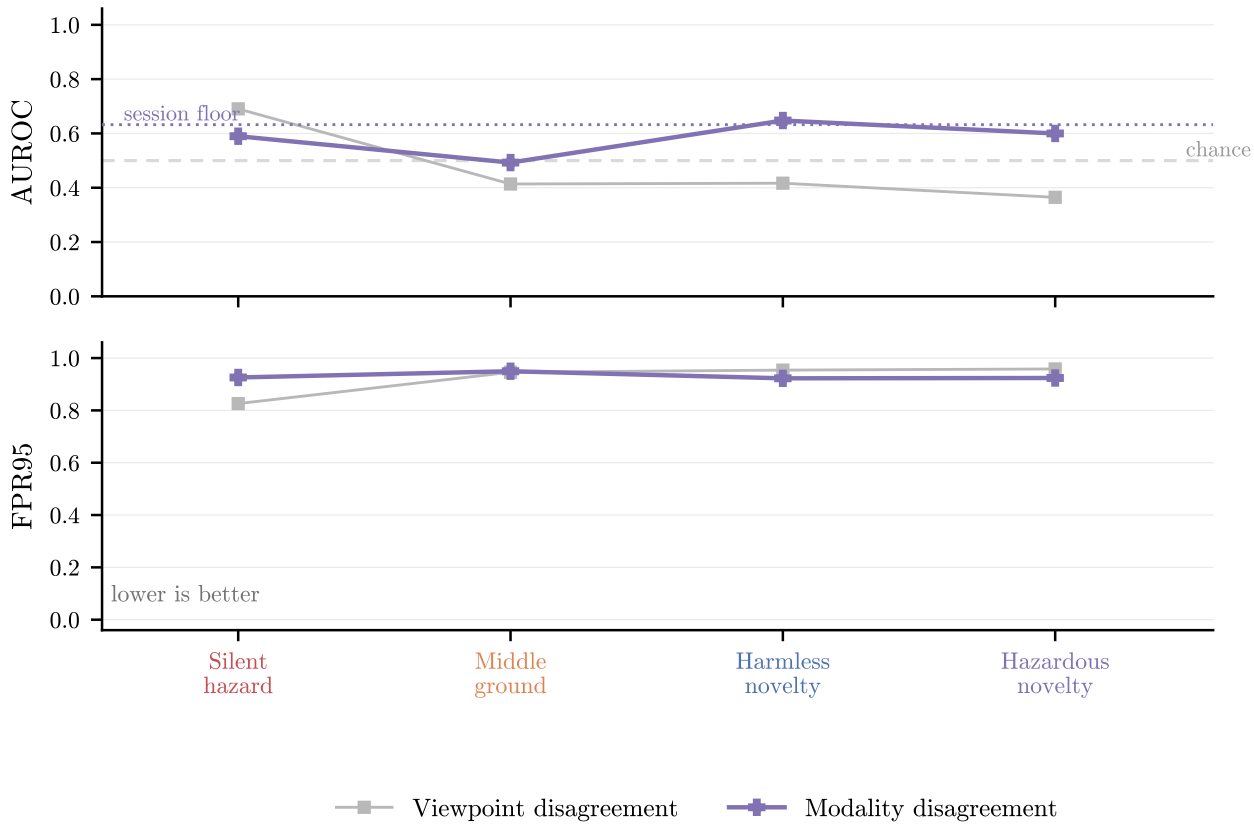
AUROC per panel, session floor 0.60

ForEnt: different robot and site, no clip in this deck.

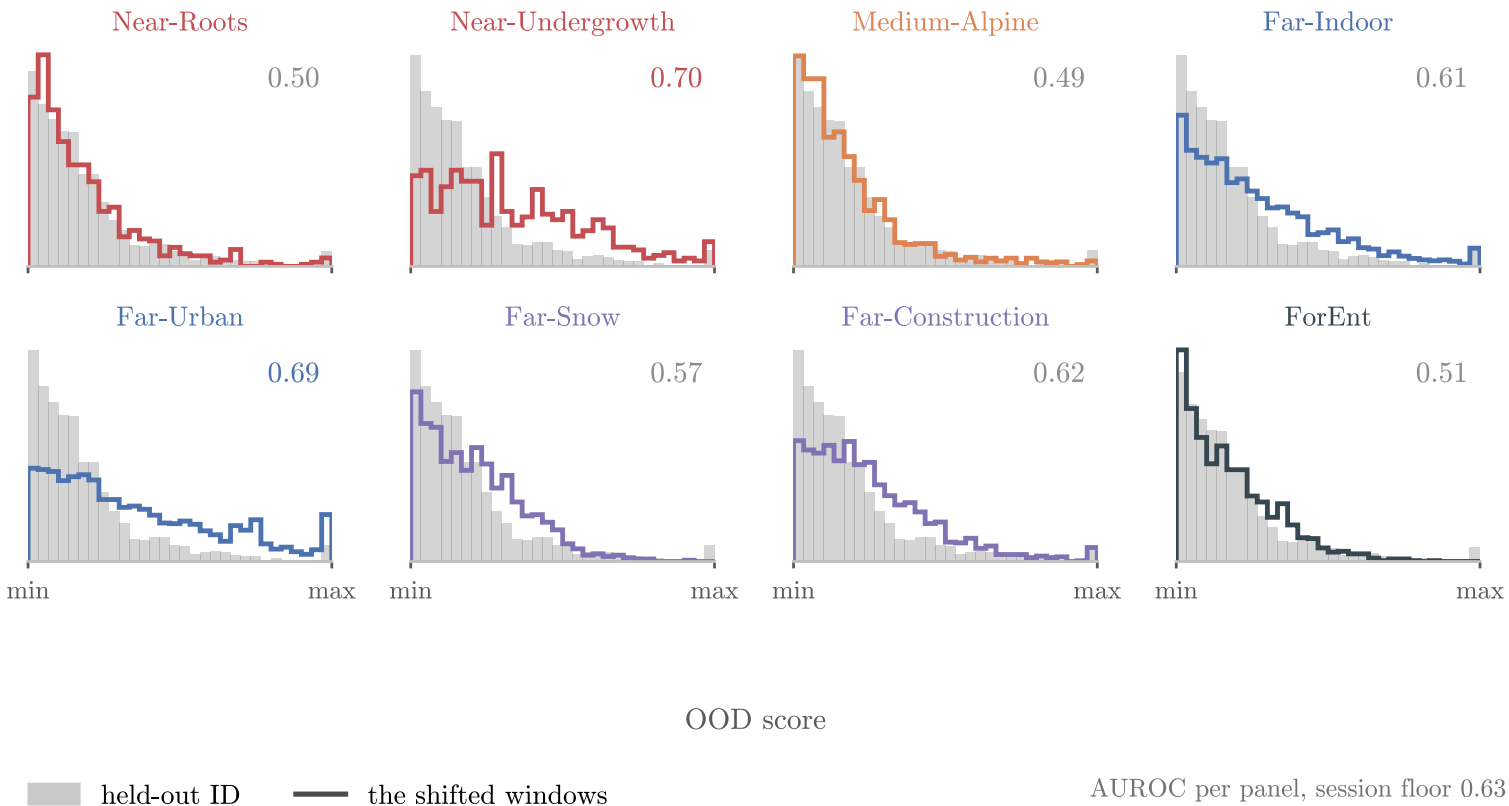
Viewpoint Disagreement - Every Tier, AUROC and FPR95



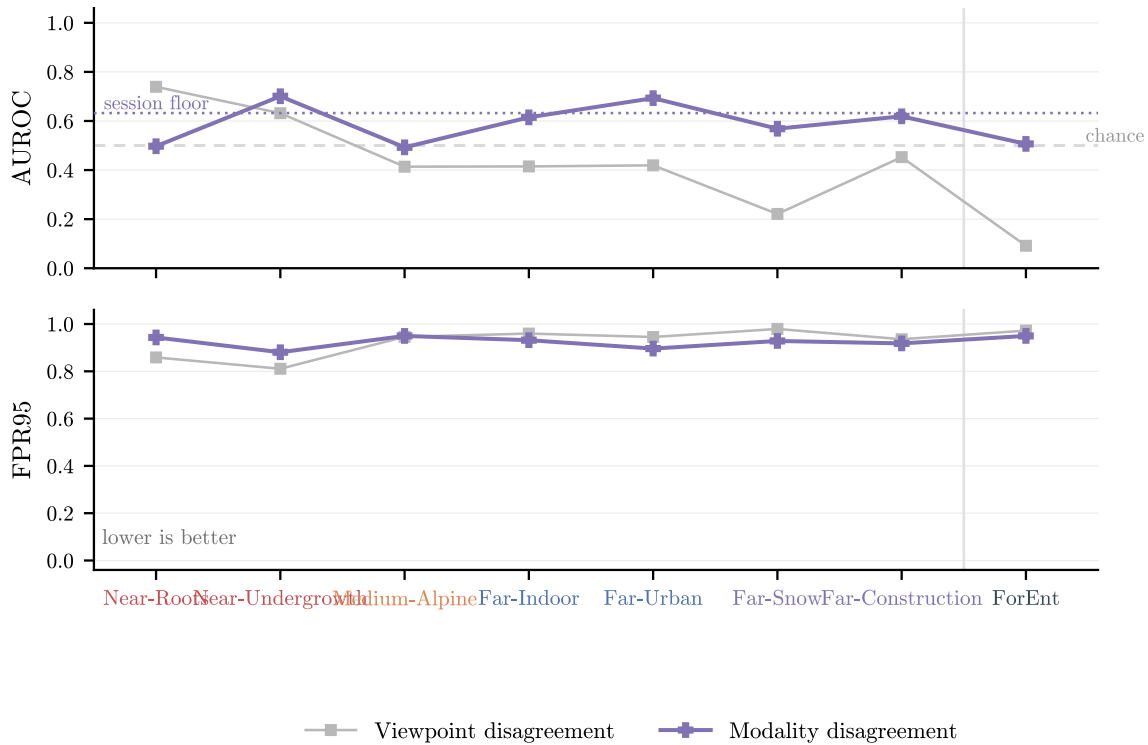
Modality Disagreement - AUROC and FPR95



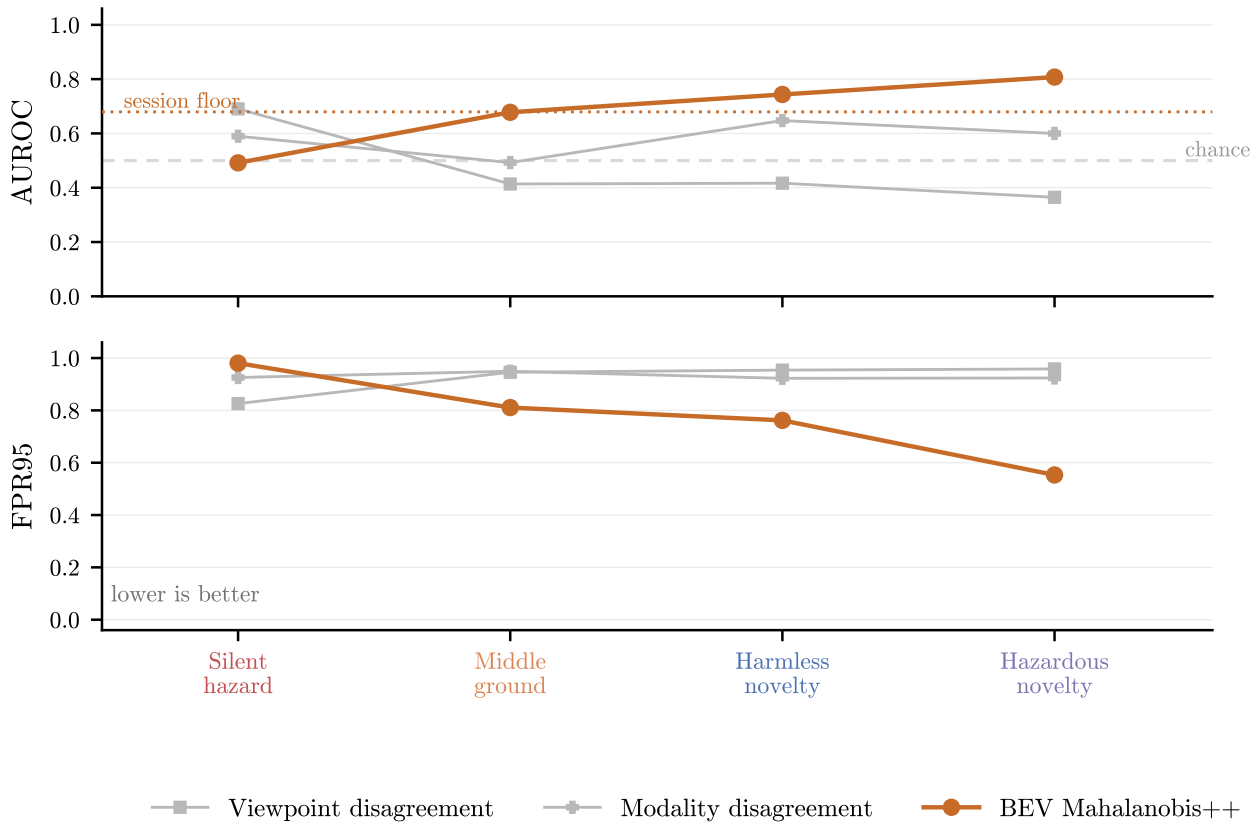
Modality Disagreement - Every Tier



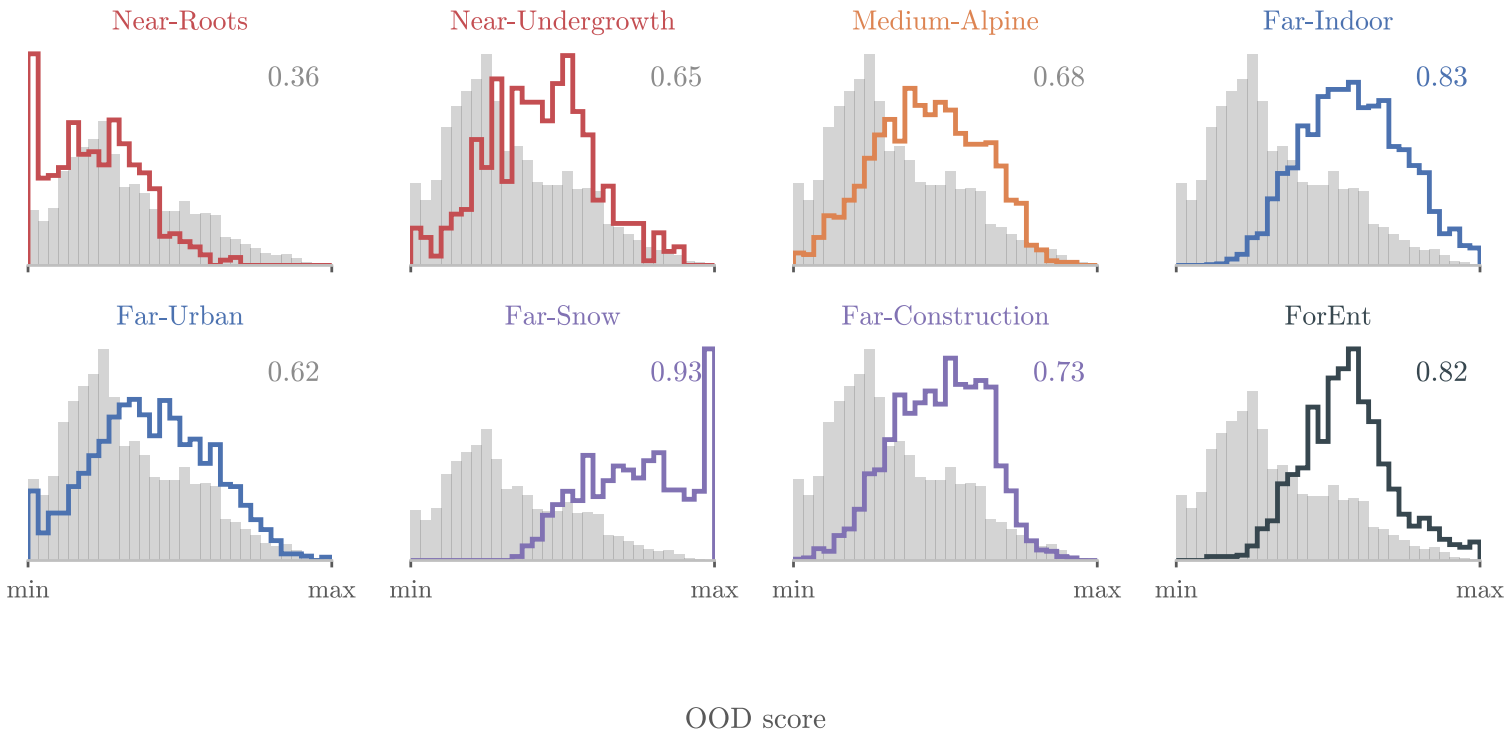
Modality Disagreement - Every Tier, AUROC and FPR95



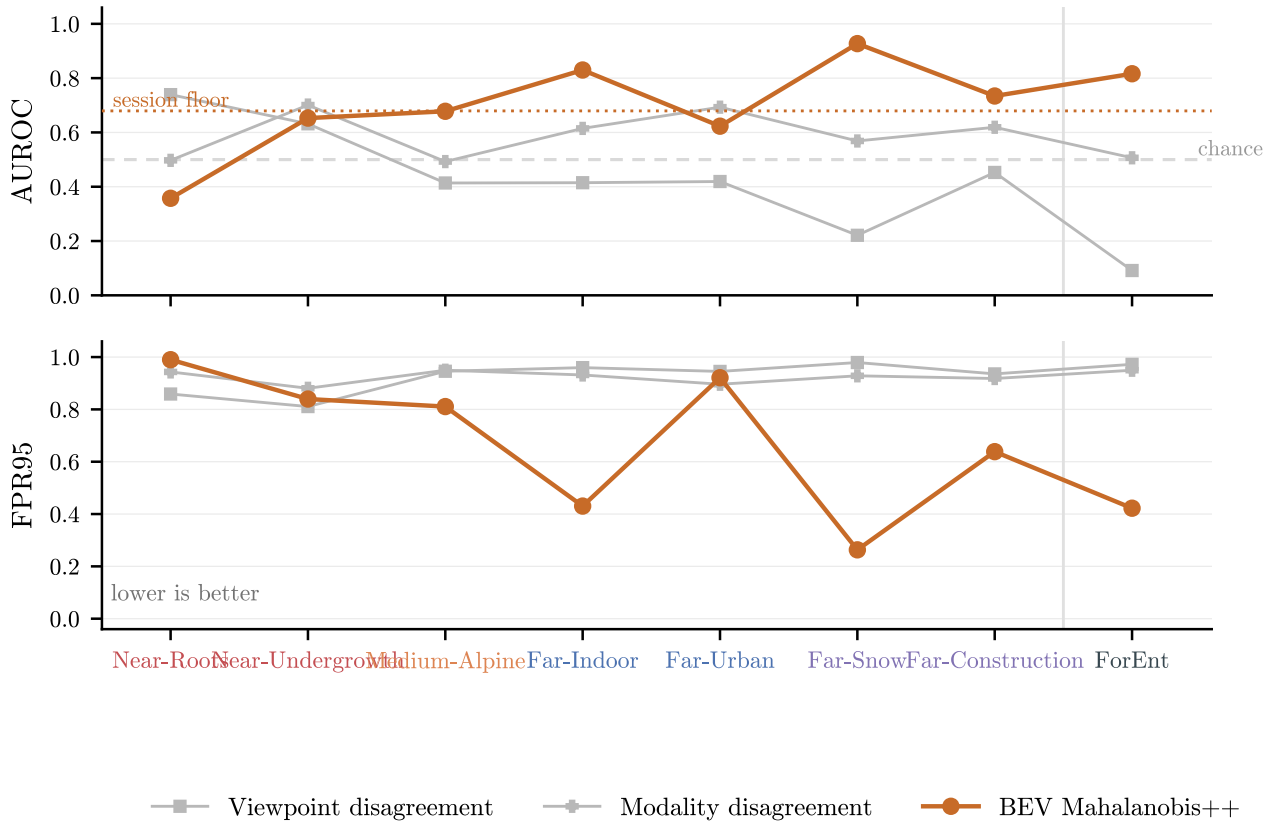
BEV Mahalanobis++ - AUROC and FPR95



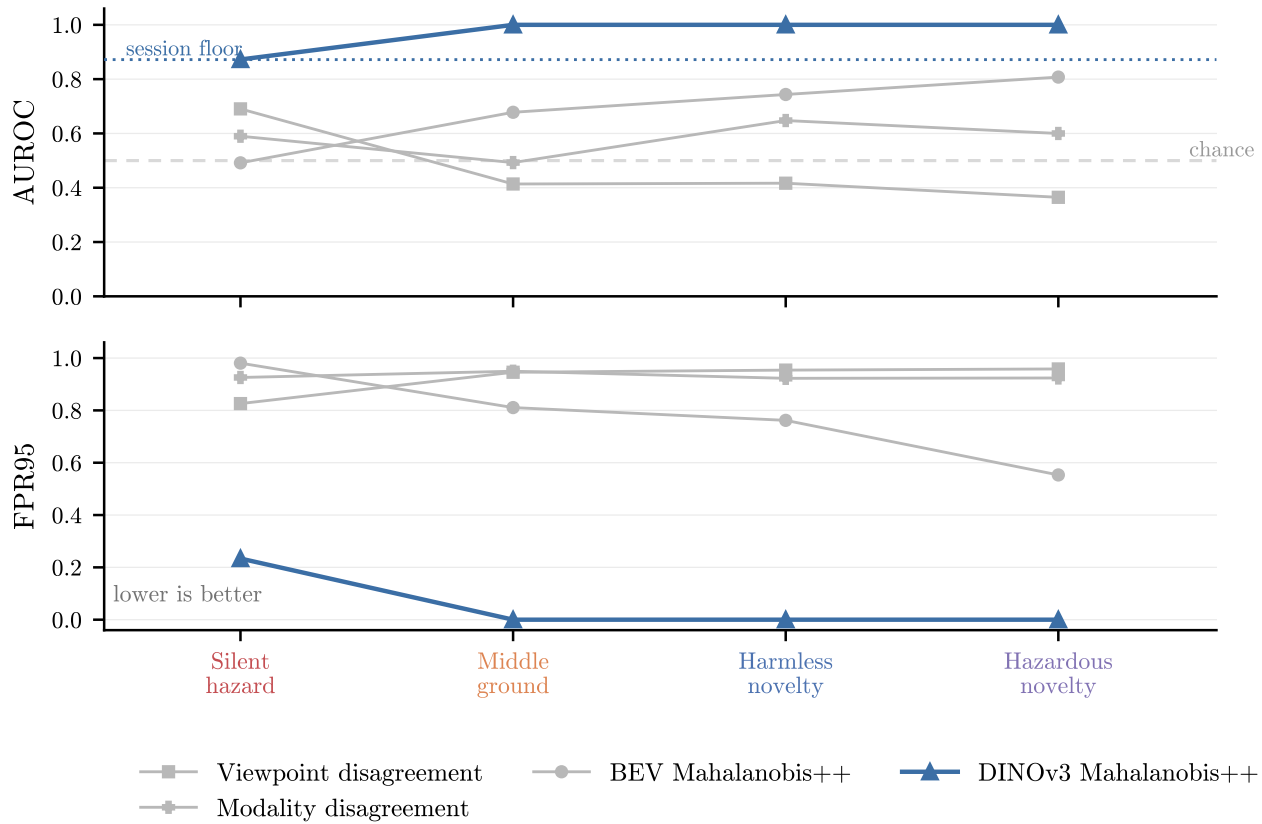
BEV Mahalanobis++ - Every Tier



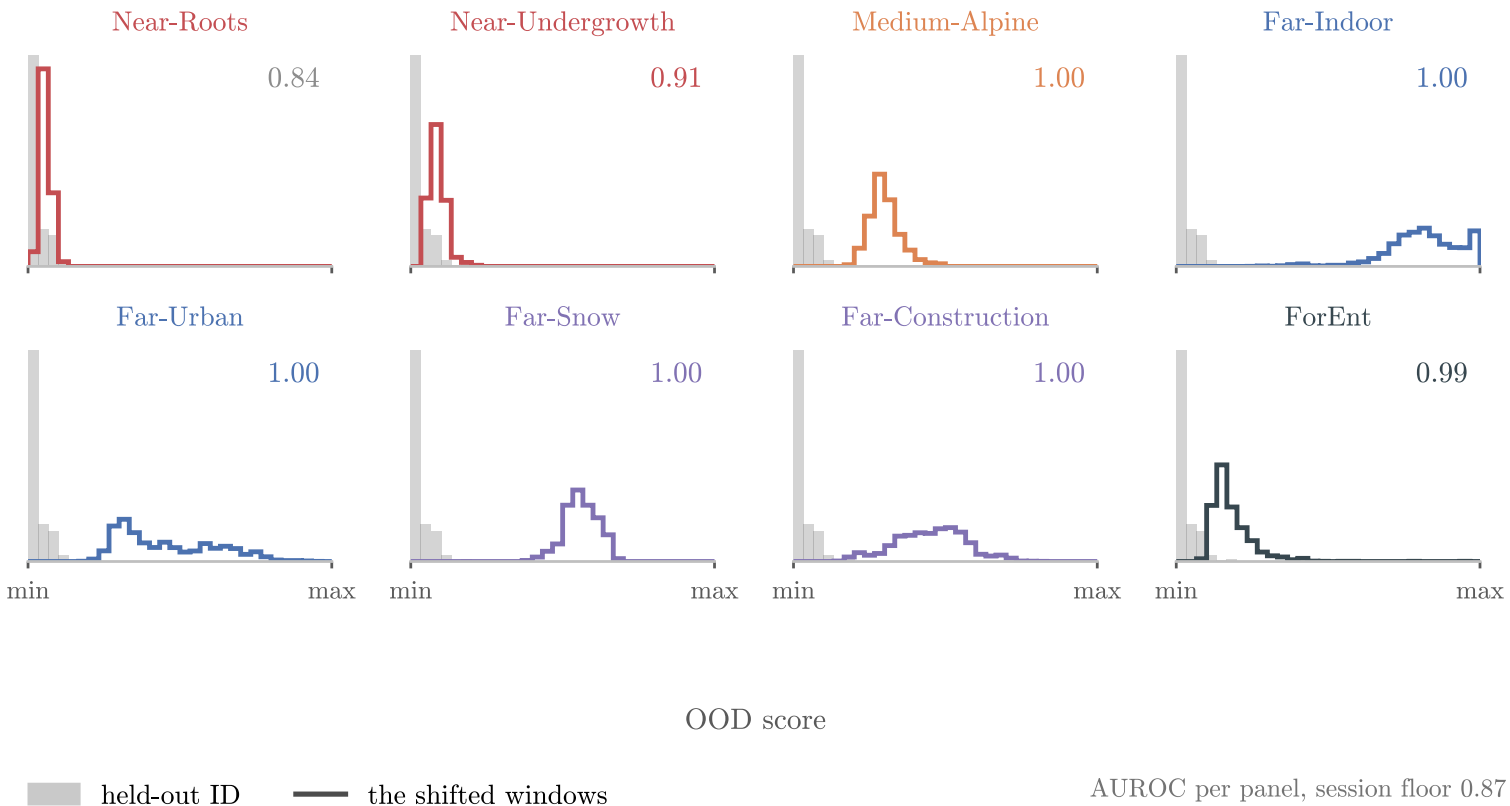
BEV Mahalanobis++ - Every Tier, AUROC and FPR95



DINOv3 Mahalanobis++ - AUROC and FPR95

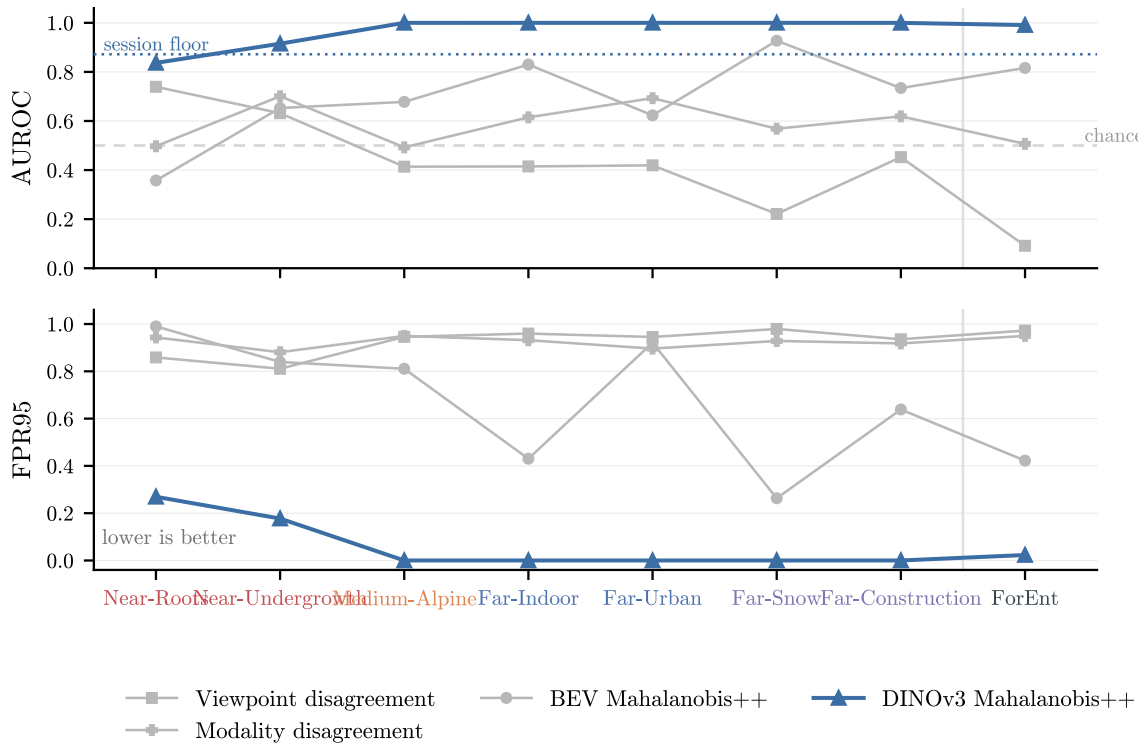


DINOv3 Mahalanobis++ - Every Tier

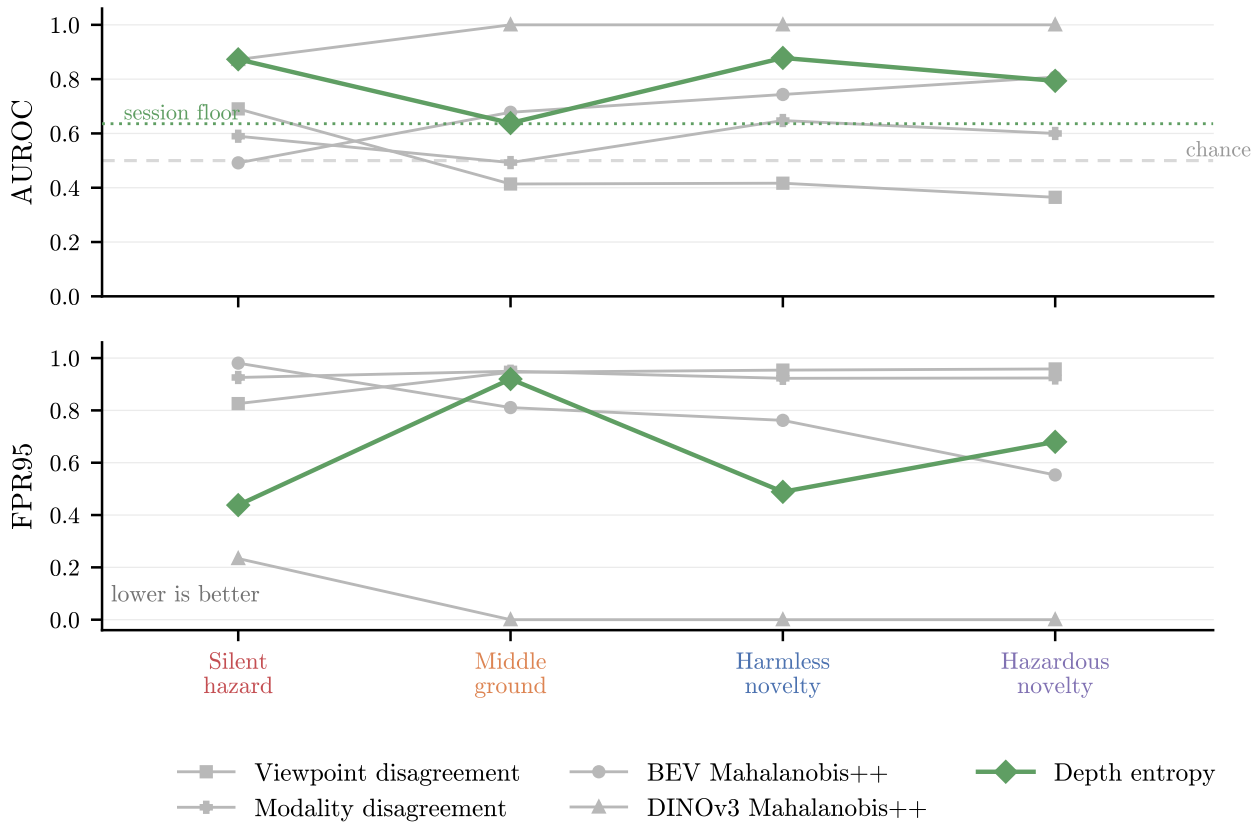


AUROC per panel, session floor 0.87

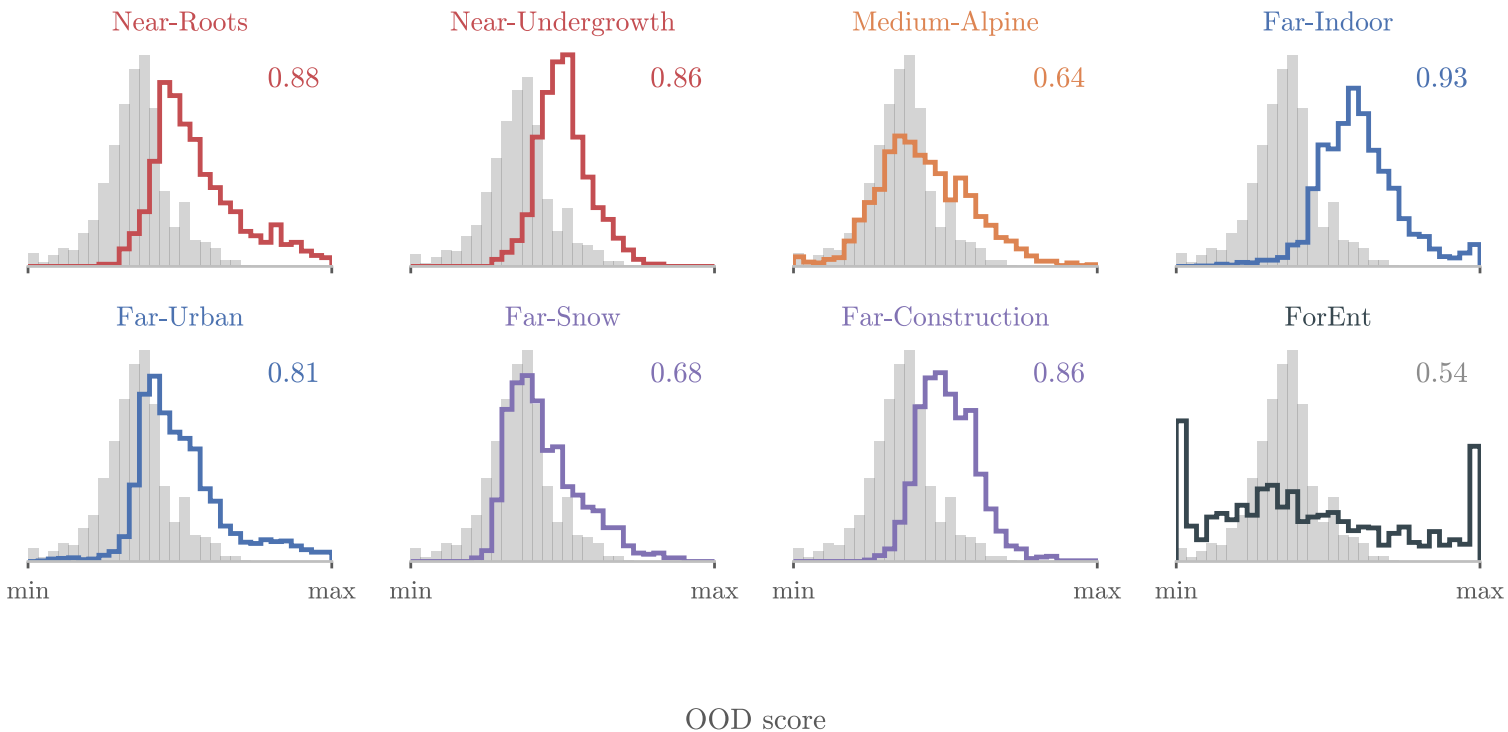
DINOv3 Mahalanobis++ - Every Tier, AUROC and FPR95



Depth Entropy - AUROC and FPR95



Depth Entropy - Every Tier



Depth Entropy - Every Tier, AUROC and FPR95